

Bhavik Kantilal Bhagat — Internship Work Logs

Internship Period: March 16, 2026 – August 31, 2026 | SMU MCDA Major Project (A00494758) | Citco Technology Management, Halifax NS | Supervisor: Kishor Deotale

Week	Dates	Task	Description	Hours	Weekly Hours
Week 1	Mar 16 – Mar 20, 2026	Onboarding & Environment Setup	- Completed Day-1 orientation covering team structure, project scope, and internship expectations with Kishor Deotale. - Attended team introduction call with co-interns Sridhar and Soundarya, and company tour. - Raised Confluence access request ticket (#14193911). - Raised JIRA access tickets (#14193674, #14193943) for the IISS project board. - Submitted BitBucket repository access request (#14196961). - Initiated AWS Core Dev access request through IDM portal. - Installed local development tools: VSCode, Git, Postman, PyCharm, Notepad++ (ticket #14195255). - Configured VSCode with Python extension, GitLens, and AWS Toolkit. - Submitted Amazon Q VSCode plugin access request (#14196961). - Requested RPA Support Agent access (#14197075).	5	13
		Environment Configuration & Architecture Study	- Configured AWS CLI with Citco's local machine connection guide for SSO-based authentication. - Tested AWS SSO login flow and verified access to citcoshareddev01 account via console. - Explored AWS Console to understand the organization structure across multiple accounts. - Navigated Confluence Innovation IT space to understand team documentation structure and naming conventions. - Reviewed Lambda function naming conventions — discovered 51 functions across rpacfs1 and IA-BPO projects. - Studied Meridian platform architecture documentation covering ECS services and containerized workflows. - Explored CitcoWorks Event Store documentation for event-driven architecture patterns. - Reviewed IA Automation Catalog to understand project registration and tracking processes. - Studied the five budget code environments (citchareddev01, citcoredev01, citcosgdev01, coretest01, coreprod01). - Explored CloudWatch console to understand existing log groups and metric namespaces.	5	
		KT session with Horace for RPA Support Queue	- Attended BPM Control Questionnaire security review meeting. - Learned about Citco's Line of Business (LoB) role structure and Active Directory integration. - Reviewed penetration testing process and Log4j vulnerability remediation history. - Learned about access logging: ELK, Kibana, Elasticsearch, TIBCO logs with 7-14 day retention. - Studied Dynatrace application monitoring for anomaly detection and alerting. - Understood UAT vs Production environment data handling policies.	3	

Week 2	Mar 23 – Mar 27, 2026	AWS Database Services Deep Dive	- Studied AWS relational databases: RDS architecture with Single-AZ and Multi-AZ deployment options. - Learned Aurora auto-scaling capabilities and cost advantages over standard RDS. - Explored Redshift data warehouse for OLAP workloads and SQL-based analytics. - Studied NoSQL options: DynamoDB schema flexibility and provisioned vs on-demand capacity modes. - Learned DocumentDB for MongoDB-compatible workloads with managed scaling. - Explored in-memory caching with ElastiCache: Redis (advanced data structures, pub/sub) vs Memcached (multi-threaded, simple). - Studied Amazon KeySpaces for Apache Cassandra-compatible wide-column workloads. - Deep dive into Neptune Graph DB: nodes, edges, Gremlin vs SPARQL query languages. - Understood Neptune use cases: social networks, fraud detection, knowledge graphs, recommendation engines. - Reviewed Amazon TimeStream for IoT time-series data with automatic data lifecycle management.	5	15
		CloudWatch & CloudFormation Study	- Studied CloudWatch CMAA cycle: Collect, Monitor, Act, Analyze. - Learned CloudWatch metrics: EC2 (1min detailed vs 5min free), custom metrics, 15-month retention. - Explored CloudWatch Logs: vended logs, published logs, on-premises agent integration. - Studied CloudWatch Alarms: metric alarms, composite alarms, SNS/Lambda/Auto Scaling actions. - Learned CloudWatch Events (EventBridge): real-time streams, rules, targets. - Completed CloudFormation IaC fundamentals: templates (YAML/JSON), stacks, change sets. - Understood CF template structure: Resources (mandatory), Parameters, Mappings, Outputs, Conditions. - Practiced writing simple CloudFormation YAML templates for S3 bucket and Lambda function resources. - Studied CloudFormation stack operations: create, update, delete, and rollback behaviors. - Explored CloudFormation change sets to understand how to preview infrastructure changes before applying.	5	
		AWS Lambda & Serverless Architecture	- Deep dive into AWS Lambda execution environment: runtime lifecycle, handler function structure. - Studied Lambda memory configuration (128MB–10GB) and CPU allocation proportional to memory. - Learned about cold starts vs warm starts and strategies to minimize latency. - Explored Lambda triggers: API Gateway, S3 events, SQS, EventBridge, CloudWatch Events. - Studied Lambda layers for sharing code and dependencies across functions. - Learned Lambda concurrency: reserved concurrency, provisioned concurrency, account limits. - Explored Lambda environment variables and AWS Secrets Manager integration for configuration. - Studied Lambda VPC configuration for accessing private resources (RDS, ElastiCache). - Reviewed Lambda logging to CloudWatch Logs and structured logging best practices. - Practiced creating a simple Lambda function with Python runtime via AWS Console.	5	

Week 3	Mar 30 – Apr 03, 2026	CTI CloudWatch Metrics & Dashboard	- Added CloudWatch metrics instrumentation to track CTI submission activities within the pipeline. - Created CloudWatch Pipeline Health Dashboard for operational visibility into system performance. - Refactored CloudWatch metrics code into a separate dedicated module for better maintainability. - Added comprehensive unit tests for metrics functionality. - Updated .gitignore to exclude virtual environment folders, pytest cache, and test artifacts. - Removed .python-version file from git tracking. - Cleaned up pycache directories previously tracked in the repository. - Merged upstream changes from origin master branch into the feature branch. - Updated dashboard to meet acceptance criteria: 1-hour granularity, 7-day display range, auto-refresh. - Added CloudWatch dashboard screenshots to support the PR review process.	6	16
		CTI CloudWatch Dashboard Log Insights	- Added technical dashboard with CloudWatch Log Insights widget for deeper system analysis. - Created CloudWatch Logs Insights queries to extract error patterns from Lambda execution logs. - Configured Log Insights widget with appropriate time ranges and query syntax for CTI pipeline logs. - Added CloudWatch Logs Insights query setup screenshots for documentation and PR review. - Removed .coverage file from git tracking. - Cleaned up extra files no longer needed from the repository. - Updated .gitignore to exclude temporary PR markdown files from version control. - Tested Log Insights queries against production log groups to validate query accuracy. - Refined dashboard layout to improve usability and information hierarchy. - Documented Log Insights query syntax and widget configuration for team knowledge sharing.	6	
		CTI Dashboard Testing & Code Quality	- Added comprehensive unit tests for CloudWatch Logs Insights query functionality. - Created unit tests for dashboard CloudFormation templates to validate deployment reliability. - Refactored code by moving imports to the top of files following Python best practices. - Suppressed Pylance warnings for cleaner code structure and improved IDE experience. - Ran full test suite — verified all 262 tests passing before PR submission. - Reviewed code coverage report to identify untested code paths. - Added type hints to new functions for improved code documentation and IDE support. - Performed code review self-check against team coding standards. - Updated README documentation with dashboard usage instructions and screenshots. - Prepared PR description with detailed summary of changes and testing performed.	4	
		CTI Technical Dashboard & Branch Merge	- Added technical dashboard with CloudWatch Log Insights widget for deeper system analysis. - Created comprehensive unit tests for CloudWatch Logs Insights query functionality. - Added unit tests for dashboard CloudFormation templates to ensure deployment reliability. - Refactored code by moving imports to the top of files and suppressing Pylance warnings. - Updated .gitignore to exclude temporary PR markdown files from version control. - Added CloudWatch Logs Insights query setup screenshots for documentation purposes. - Removed extra files no longer needed to clean up the repository. - Removed .coverage file from git tracking. - Updated the dashboard to meet acceptance criteria: 1-hour granularity, 7-day display range, auto-refresh. - Added CloudWatch dashboard screenshots to support the PR review process.	7	

Week 4	Apr 06 – Apr 10, 2026	IA Lambda Monitoring Dashboard	- Created static CloudWatch dashboard IA-Lambda-Monitoring-Dashboard. - Added filter info widget at the top with usage instructions for team members. - Implemented overview widgets displaying Total Invocations, Errors, and Throttles across all 51 Lambda functions. - Added error analysis widgets: Error Count and Invocations vs Errors bar charts for trend analysis. - Created error investigation widget using CloudWatch Logs Insights for root cause analysis. - Added duration metrics widgets: Average, Maximum, and P99 latency values. - Implemented cold starts, concurrency, throttling, and memory utilization monitoring widgets. - Designed modular YAML widget architecture for easy maintenance and extensibility. - Added comprehensive unit tests — 21 tests passing. - Fixed dashboard builder to validate text widgets with markdown property correctly.	7	18
		Lambda Dashboard PR Review & Refinement	- Addressed PR review feedback on Lambda Monitoring Dashboard from senior team members. - Refined dashboard widget layout based on usability suggestions from code review. - Added additional test cases for edge cases identified during PR review. - Created IISS-461 subtask for Confluence Lambda inventory documentation. - Published Confluence page with full Lambda inventory covering all 51 functions across rpacfs1 and IA-BPO. - Documented Lambda trigger mappings (S3, SQS, EventBridge, API Gateway) for each function. - Created budget code table mapping Lambda functions to cost allocation tags. - Added function-level documentation: memory configuration, timeout settings, and runtime versions. - Verified Lambda inventory accuracy by cross-referencing AWS Console with Confluence documentation. - Updated team runbook with links to the new Lambda monitoring dashboard and inventory page.	4	
Week 5	Apr 13 – Apr 17, 2026	Multi-Dashboard Architecture Refactor	- Refactored the monitoring solution to a multi-dashboard architecture for better organization and maintainability. - Implemented explicit Lambda list filtering to ensure dashboards display only relevant functions. - Added error investigation capabilities to help with root cause analysis of Lambda failures. - Fixed issues with the health dashboard to improve accuracy of displayed metrics. - Resolved bar chart rendering issues in the dashboard widgets. - Separated dashboard concerns into distinct views for better user experience and focused monitoring.	8	42.5
		Training Week Day 2	- Attended Citco corporate onboarding and compliance training sessions. - Completed mandatory security awareness training modules. - Reviewed IA team documentation and Confluence knowledge base. - Met with team members for introductions and role overview.	8	
		Training Week Day 3	- Continued Citco corporate training modules. - AWS environment walkthrough with Horace (Ho Hoi Leung). - Introduction to IA platform architecture and service topology. - Reviewed existing CloudWatch dashboards and monitoring approach.	8	

		Training Week Day 4	- RPA platform overview: Blue Prism / UiPath architecture KT. - Event Store API and logging standards training — patterns will be used in dashboard work. - IA Support Model walkthrough: Follow-the-Sun coverage, incident response, escalation paths. - Reviewed RPA support responsibilities for future rotations.	8	
		Development & Implementation	- Continued development work and implementation tasks.	8	
		Meetings & Standups	Meetings this week (4 meetings, 2.5h): - SRE Team Standup (30 min) - IA Team Sync (60 min) - 1:1 with Manager (30 min) - Knowledge Transfer (30 min)	2.5	
		Centralized CloudWatch Dashboard Improvements	- Added a monitoring export Lambda function with daily cron scheduling and automatic S3 upload capability. - Implemented S3 upload functionality for monitoring reports and resource lists. - Renamed lambda_registry_api to resources_monitoring_api for clearer naming reflecting broader resource coverage. - Added S3 Lambda list fallback mechanism to ensure dashboard works even when API is unavailable. - Created an export API endpoint with daily granularity for data retrieval. - Fixed export filename format to use on_{requested}_from_{start}_to_{end} naming convention. - Converted the registry API from FastAPI to Flask with Swagger documentation for better integration. - Updated the dashboard to pull Lambda data from the API instead of static files.	8	
		Lambda Trigger Detection, API Refactor & Test Overhaul	- Added Lambda trigger source detection to identify whether functions are triggered by SQS, API Gateway, or EventBridge/Scheduled events. - Implemented detection logic using list_event_source_mappings, get_policy, and describe_rule APIs. - Fixed S3 upload crash on the export endpoint and renamed the route to /export/lambdas for clarity. - Added type hints, return types, and docstrings across both packages and fixed ruff/pyright warnings. - Refactored resources_monitoring_api by extracting namespaces, models, and parsers into dedicated modules. - Overhauled the test suite for both packages — 262 tests passing with pyright clean status. - Updated project configuration for pytest, coverage, and test dependencies. - Updated the resources_monitoring_api README with production server commands and refactored structure documentation.	8	
		EC2/ECS/EKS Resource Discovery & API Consolidation	- Added EC2, ECS, and EKS resource discovery endpoints to resources_monitoring_api for broader infrastructure visibility. - Refactored the API by consolidating all resource endpoints under the /api/v1/resources/ path for consistent routing.	4	
		Versioned S3 Export & DEV Bucket Fix	- Implemented versioned Lambda list saving to S3 alongside the latest version for historical tracking. - Changed the DEV S3 bucket to dev-citco-ia-checklist-service-documents for proper environment alignment.	4	

Week 6	Apr 20 – Apr 24, 2026	API Swagger Redirect, Registry Refactor & Package Structure	- Updated the monitoring export Lambda to use the new DEV S3 bucket for proper environment configuration. - Added redirect from root URL to Swagger UI at /docs for better API discoverability. - Fixed the redirect to properly route GET / requests to the Swagger documentation. - Refactored the registry by extracting it into a dedicated package with per-resource sub-modules. - Added pyproject.toml per package and split dev/test dependencies for better dependency management. - Created a CONTRIBUTING.md file to document contribution guidelines. - Removed stale files from resources_monitoring_api and fixed ruff UP/isort findings. - Removed redundant region/account_id fields from resource item models to simplify the data structure.	4	42.5
		EC2 Registry Module	- Added availability_zone and tags fields to EC2 instance response for richer resource metadata. - Created a standalone ec2_registry.py module with session-based API for EC2 resource discovery.	4	
		ia-resources-monitoring Repo Setup, Code Quality & CLI Entry Point	- Set up new ia-resources-monitoring repository with initial docs, requirements, pyproject.toml, and pre-commit configuration. - Added limitations documentation and dashboard screenshots. - Created CONTRIBUTING.md, lambda-monitoring-export CloudFormation template, and output folder structure. - Implemented code quality improvements: ruff fixes, mypy config, deptry, pre-commit configs, Python 3.14 support. - Updated technical guide and limitations documentation for the full platform (Lambda + EC2 + API). - Added boto3-stubs and pyright configuration to fix type errors and tighten mypy overrides. - Replaced unsafe TypedDict bracket access with .get() and fixed except syntax in routes.py. - Created ia_paths.py module and refactored all sys.path.insert calls to use package imports.	2	
		resources_monitoring_api Migration	- Migrated resources_monitoring_api package to the new ia-resources-monitoring repository.	2	
		ia_lambda_monitoring_dashboard Migration	- Migrated ia_lambda_monitoring_dashboard package to the centralized ia-resources-monitoring repository.	2	
		EC2 Dashboard Migration & rpacfs1 EC2 Work	- Migrated ia_ec2_monitoring_dashboard package to the centralized ia-resources-monitoring repository. - Added ia_ec2_monitoring_dashboard package with navigation bar in rpacfs1 repo. - Aligned EC2 S3 storage to shared checklist bucket using the same pattern as Lambda. - Added GET /api/v1/export/ec2 endpoint for EC2 resource exports. - Extracted exporters/ package from routes.py for better code organization. - Moved cais-pe-dashboard CloudFormation templates to project root for easier access.	2	
		Meetings & Standups	Meetings this week (4 meetings, 2.5h): - SRE Team Standup (30 min) - IA Team Sync (60 min) - 1:1 with Manager (30 min) - Sprint Planning (60 min)	2.5	

Week 7	Apr 27 – May 01, 2026	ECS Monitoring Dashboard	- Added the <code>ia_ecs_monitoring_dashboard</code> package with complete dashboards, widgets, tests, CloudFormation templates, export functionality, and deploy support. - Fixed the preview functionality to use resolved output showing text widget titles and replacing placeholders correctly. - Fixed health widgets by removing SQL arithmetic operations that were causing issues. - Replaced log widget with text placeholder for better reliability. - Updated to use native <code>PendingTaskCount</code> metric instead of calculated values.	8	42.5
		src/api Package Restructure & Save-Queries CLI	- Added <code>--delete</code> flag to <code>save-queries</code> CLI for CloudWatch Logs Insights query management. - Restructured the <code>ia-resources-monitoring</code> project into <code>src/</code> and <code>api/</code> packages for better organization.	1.5	
		EKS Monitoring Dashboard	- Completed the EKS monitoring dashboard with full functionality. - Fixed S3 export bucket configuration and added EKS paths to shared config. - Fixed Metrics Insights SQL quoting by using single quotes in <code>IN</code> clause. - Fixed SQL quoting issue by replacing cluster names before JSON serialization. - Split EKS widget YAMLS into 1 file per row for easier maintenance. - Replaced SQL expressions with placeholder-based metrics in EKS widgets for reliability. - Moved error investigation from health dashboard to overview and added EKS saved query.	1.5	
		ECS Widget YAML Split	- Split ECS widget YAMLS into 1 file per row following the <code>src/</code> structure pattern.	1.5	
		Output Structure, S3 Prefixes & Dashboard Builder Updates	- Added navigation and header widgets to all ECS and EKS dashboards for consistent UX. - Consolidated output folders to use <code>_lists</code> suffix for consistent naming. - Fixed S3 export buckets and report folder structure. - Restructured output to use <code>{ENV}/resources/</code> and <code>{ENV}/reports/</code> paths. - Removed old output folders and migrated data to the new structure. - Added per-resource S3 resource prefixes for better organization. - Updated dashboard builders to use <code>{ENV}/resources/</code> paths.	1.5	
		cais-pe Dashboard Pipeline & rpacfs1 Cleanup	- Removed monitoring packages from <code>rpacfs1</code> repo — fully migrated to <code>ia-resources-monitoring</code>. - Removed docs/screenshots from <code>rpacfs1</code> — migrated to centralized repo. - Made <code>cais-pe</code> dashboards fully environment-aware for UAT/PROD deployment. - Added UAT/PROD deployment runbook for <code>cais-pe</code> dashboards. - Added <code>cais-pe</code> dashboards as nested stacks in <code>main-template.yaml</code>. - Updated <code>AlarmEMailList</code> in <code>dev.json</code> to match UAT/PROD configuration. - Removed <code>DASHBOARD_DEPLOYMENT_RUNBOOK.md</code> — dashboards now deployed via pipeline. - Added <code>@log</code> field to error investigation query in technical dashboard for better log filtering.	1.5	
		Logs Insights Optimisation, Saved Queries & Code Quality	- Removed old output folder creation from <code>lambda</code> exporter and associated tests. - Updated all markdown files to reflect current state: 4 resources, 12 dashboards, EKS support, correct paths. - Updated PR template for release/SRE to master merges. - Resolved <code>mypy</code>, <code>pyright</code>, and <code>deprity</code> type errors across <code>api</code> and <code>src</code> packages. - Added <code>pyright</code> to <code>requirements-dev.txt</code> for type checking support. - Formatted EKS test files with <code>ruff</code> for consistent code style. - Saved Logs Insights queries under the <code>Intelligent-Automation</code> folder for better organization. - Fixed delete functionality to handle both folder-prefixed and legacy query names.	8	

		EC2 Dashboard Polish	- Fixed EC2 dashboard discovery to use AppName fallback for instances without standard naming. - Fixed EC2 dashboard layout and added missing performance widgets. - Reordered overview rows and grouped performance widgets by category for better organization. - Redesigned the EC2 performance dashboard and fixed header hierarchy issues. - Increased EC2 dashboard header height from 4 to 5 for better readability. - Added trailing spaces for markdown line breaks in EC2 and overview headers to ensure proper rendering. - Fixed header height to 4, added Credit Balance widget to health dashboard, and added annotations and AVG line to performance dashboard.	2.75	
		Dashboard Header Standardisation	- Standardized dashboard headers across all resources and excluded them from the trailing-whitespace pre-commit hook.	2.75	
		ECS Reservation Widgets Restore	- Restored CPU/Memory reservation widgets that were previously removed.	2.75	
		ECS Service Inventory Table	- Replaced ECS reservation widgets with a service inventory table for better visibility.	2	
		EKS Dashboard Finalisation & Merge	- Switched EKS dashboard to use AWS/EKS control plane metrics for accurate monitoring. - Fixed EKS Overview layout for proper widget alignment. - Fixed Cluster Status empty state markdown to display correctly when no clusters exist. - Added save_eks_list exporter and updated local fallback mechanism. - Split overview YAML into 1 file per row and added ## widget titles for consistency. - Removed ## from metric widget titles to fix rendering issues. - Merged the EKS monitoring dashboard feature branch into release/SRE. - Increased EKS test coverage to 81% for better code reliability.	2	
		Structured Logging & Test Coverage	- Added .gitattributes for line ending normalization across different platforms. - Replaced print statements and disable_warnings with structured logging in the API package. - Implemented structured logging across EKS, ECS, EC2, and Lambda dashboard packages. - Merged the structured logging refactor into release/SRE. - Increased test coverage from 67% → 75% (754 tests), 75% → 77% (784 tests), and hit 81% (830 tests). - Merged test coverage improvements into release/SRE.	2	
		EC2 Dashboard Improvements	- Improved EC2 dashboard with Fleet Status, EBS metrics, and removed Fleet Avg widget.	2.25	
		Meetings & Standups	Meetings this week (4 meetings, 2.5h): - Tech Talk AI Strategy (30 min) - Configuration Service DEMO (30 min) - Container Insights Cost Discussion for Dashboard (30 min) - Daily Standup (30 min)	2.5	

Week 8	May 04 – May 08, 2026	RPA Production Support Shift	- Reviewed Confluence shift handover log from APAC team. - Monitored SD Portal queue and Teams channels for incoming tickets. - Handled routine queue item retries and process status inquiries. - Updated ticket statuses and prepared end-of-shift handoff notes. - Reviewed incoming tickets and APAC handoff notes. - Investigated MESO process status inquiries — checked CloudWatch logs by Run ID. - Handled FX Closeout retrigger requests — verified Daily565 folder, archived old reports. - Updated ticket documentation with actions taken and next steps. - Monitored UiPath Orchestrator for job statuses and queue health. - Handled Corporate Actions ticket — updated file masks in BPSHare templates for new client onboarding. - Investigated Auto Rec Download config request — backed up and edited gridview.py and config.ini. - Checked MESO DB queries for client report status.	40	40
		EC2 Overview Widget Layout Fix	- Adjusted widget heights and y-offsets for the EC2 overview dashboard for better visual layout.		
		BudgetCode-First Discovery & ECS Filter	- Implemented BudgetCode-first discovery with AppName-fallback for flexible resource identification. - Added input sanitization to protect against invalid query parameters. - Added ECS budget_code filter for cost-based resource filtering. - Fixed test mocks to align with the new filtering logic.		
		MERIDIAN ECS Cluster-Based Fallback	- Added cluster-based fallback for ECS service discovery when tag-based discovery fails — required for MERIDIAN platform where 5 of 6 ECS clusters are untagged and need name-based discovery.		
		Lambda Dashboard Budget Summary & API Fixes	- Removed the TopK dashboard variable widget from Lambda dashboards. - Merged upstream changes from release/SRE into the Lambda monitoring dashboard feature branch. - Fixed API by enabling threaded mode, fixing URL path, and increasing timeout for better reliability. - Added budget code summary widget to all Lambda dashboards for cost visibility. - Implemented clickable Lambda inventory at top with budget summary at bottom for improved dashboard layout.		
		Lambda Dashboard Widget Position Fix	- Swapped widget positions in Lambda dashboards placing summary at top and inventory at bottom for better user experience.		
		Full Grafana Infrastructure Deploy	- Enforced coding standards including uv package manager, JavaDoc-style docstrings, and strict types. - Implemented Human-In-the-Loop (HIL) gate with trigger rules, blocking wait mechanism, and audit trail. - Implemented orchestrator routing, agent registry, and task models for multi-agent coordination. - Added CloudFormation template for Grafana workspace with associated IAM role. - Created Lambda custom resource for Grafana API provisioning to automate dashboard setup. - Added Grafana dashboard JSON rows 1-5 covering Health, SQS, Lambda, and ECS monitoring. - Added Grafana dashboard rows 6-10 covering MSK, DocumentDB, Neptune, OpenSearch, and Alerts. - Created CloudWatch alarms CFN template with all 17 alerting rules.	8	

Week 9	May 11 – May 15, 2026	Documentation Suite, 99%→100% Coverage & Release Merges	- Merged IISS-487 Grafana dashboard, alarms, CFN, pipeline, deploy scripts, and SSL fix into release. - Added release notes for IISS-487 SRE observability milestone. - Fixed .gitignore UTF-16LE corruption by rewriting the file in UTF-8. - Removed SSL setup script and certifi dependency — using Windows system cert store instead. - Restructured requirements: prod=base only, dev=test+tooling, agents=local only. - Rewrote installation guide with pip-first approach and role-based requirements. - Added architecture document, architecture diagrams (PNG), and process flow document. - Added permission fixes, IAM workarounds, and bootstrap transparency note for CFN execution role.	8	42.5
		SQS Panel Fixes, Grafana 12.4 Upgrade & Pipeline Refactor	- Fixed Queue Depth, Messages In-Flight, Age of Oldest Message, and DLQ Depth panels with per-queue breakdown and QueueName legends. - Achieved 100% test coverage on Row 2 SQS panels. - Bumped ProvisionerVersion to 9 to deploy Row 2 SQS panel fixes. - Set default time range to now-5m with 1 minute refresh; bumped ProvisionerVersion to 10. - Upgraded Grafana from 10.4 to 12.4 and bumped schemaVersion to 41. - Added check_grafana_secret preflight and iam:CreateServiceLinkedRole permission for Grafana 12. - Split test_deployment_scripts into separate test files for deploy, smoke_test, and package_lambda. - Added --pipeline flag to deploy.py and deleted deploy_pipeline.ps1/.sh.	4	
		ia_ebinder_services: Dockerfile Python Upgrade	- Updated Dockerfile in ia_ebinder_services to use python3.13 image instead of 3.12.	4	
		23-Agent Architecture, Bootstrap Flow, SQS Panels & Auto-sync .env	- Synced feature/IISS-487-pipeline with release/IISS-487-sre-observability multiple times overnight. - Added git_expert and grafana_query_expert agents with docs/ knowledge base. - Consolidated steering to .kiro/, set line-length to 120, cleaned up .amazonq. - Grouped 23 agents into infra/grafana/development/workflow/safety categories. - Added amg_deployment_expert and amg_observability_expert agent skills. - Added cloudwatch_alarms_expert agent for alarm design, thresholds, and tuning. - Enforced separation of duties and removed circular defer chains across all agent skills. - Added delegation DAG mermaid diagram to agents README.	8	
		Lambda Panels, \$platform Variable & SQS SQL Migration	- Overhauled the developer-guide by removing manual AWS CLI steps for a cleaner workflow. - Fixed SQS panels with Metric Insights GROUP BY without LIMIT, DLQ timeseries, and turned off gauge threshold labels. - Fixed SQS row with Metric Insights SQL using GROUP BY, \$queue variable, and DLQ filter. - Added \$function variable to all Lambda panels and updated SQS Health gauge thresholds. - Fixed Concurrent Executions panel with per-function breakdown mirroring the SQS format. - Fixed Error Rate panel with Metric Insights SQL and math for per-function error percentage. - Split Duration panel into p50 and p99 panels with per-function breakdown. - Fixed Lambda row with all panels using auto period, Throttles as timeseries, noValue handling, and Error Rate labels.	8	

		Meetings & Standups	Meetings this week (8 meetings, 2.5h): - MAS - Infrastructure Monitoring Review (60 min) - Daily Standup (30 min) - Daily Standup (30 min) - UiPath Sharepoint API Library - Overview (60 min) - Daily Standup (30 min) - CONFIG SERVICE DEMO - NA (30 min) - Daily Standup (30 min) - [Demo] RPA Support Agent (30 min)	2.5	
Week 10	May 18 – May 22, 2026	Rows 5-8 SQL Migration, \$platform Variable, Collapsed Rows & Docs	- Migrated Row 5 ECS APIs to SQL with per-target-group breakdown; split into p50/p95/p99 panels. - Added \$platform variable filter to Row 5 ECS APIs and shifted rows 06-10 gridPos. - Documented platform filter ALB limitation, ALB/ECS relationship, mermaid flowchart for Row 4/5, and ECS/ALB restaurant analogy. - Consolidated limitations into known-limitations.md covering ALB filter, percentile, and SQL query sections. - Changed default time range from 5d to 24h; deployed hotfix provisioner v87. - Implemented auto period with collapsed rows — provisioner v91. - Migrated Row 6 MSK, Row 7 DocumentDB, and Row 8 Neptune to SQL with per-instance/per-broker breakdown. - Added renameByRegex to all rows; separated Queue Depth from DLQ Depth filters in Row 2 SQS.	8	38
		Rows 7-9 SQL Migration, Aurora/RDS Row, SQS Inventory & Agent Roles	- Added DBCluster vs DBInstance explanation to concepts-explained documentation. - Added tag-based filtering notes to workflow steering document. - Migrated Row 9 OpenSearch to SQL with per-domain breakdown. - Merged feature/IISS-487-DocumentDB into release — completing Row 7 DocumentDB SQL migration. - Bumped provisioner version after DocumentDB deploy. - Added merge=ours strategy for grafana-workspace.yaml to prevent provisioner version conflicts. - Added friendly no-data message to Row 5 HTTP 5xx/4xx panels; applied noDataText and noValue='--' across all panels in rows 2-9. - Removed filterFieldsByName from Row 5 ALB panels (ALB names don't contain 'meridian').	8	
		ECS Row 4, \$platform Variable, SQS Health & Agentic AI Test Fixes	- Updated SQS panels to use Metric Insights SQL with filterFieldsByName and fixed allValue for meridian filter. - Added \$platform variable for multi-platform filtering using filterFieldsByName. - Fixed p50/p99 panels by adding matchExact false, period 300, and \$platform variable filter. - Implemented SQS Health stat panel with sum calculation, platform filter, and health overview improvements. - Changed SQS Health gauge to use MAX instead of SUM; bumped provisioner to v91. - Fixed Row 4 ECS with SQL migration, per-service breakdown, and corrected Panel 4 metric. - Removed SCHEMA() syntax from Row 4 ECS causing query limit error — switched to FROM namespace. - Split Row 4 ECS multi-query panels, removed blank labels, bumped provisioner to v71.	8	

		Agentic AI Test Suite Completion, Health Overview & SSL Fix	- Updated SQS row tests for new SQL expression format and added cfnlint API preflight checks. - Wrapped all CloudwatchAlarmsExpert instantiations in mock_aws for proper test isolation. - Eliminated all remaining hanging tests across the test suite. - Fixed mocking of Tester and CloudArchitect subprocesses in test_role_stub_init_and_run. - Deployed and bumped ProvisionerVersion to 352, then to 355. - Added corporate CA certificate setup script; replaced PowerShell cert setup with cross-platform Python script. - Clarified SSL issue is VPN vs office network. - Added SQS Health stat panel showing sum of meridian queue maximums with 80/100 alert thresholds.	8	
		Meetings & Standups	Meetings this week (9 meetings, 6.0h): - Daily Standup (30 min) - UiPATH - VPL PRICING (30 min) - Daily Standup (30 min) - Tech Talk AI Use Case Framework overview (30 min) - Daily Standup (30 min) - HFX Town Hall (105 min) - SRE Weekly Call (30 min) - Daily Standup (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min)	6	
		Health Panels, S3 Dashboard Storage Migration & Performance Overview	- Added DocumentDB, Neptune DB, RDS/Aurora, and OpenSearch health panels with CPU 60/80% thresholds. - OpenSearch health panel shows Green/Degraded status based on ClusterStatus.green metric. - Enhanced ECS metrics to show CPU/Memory usage MAX per service ordered by MAX() DESC. - Migrated dashboard storage from AWS Secrets Manager to S3 — removes the 65KB payload constraint. - Added ServerSideEncryption=AES256 to S3 uploads to comply with CITCO-S3ForceEncryptNoPublic policy. - Added s3:ListBucket permission to Lambda IAM role for NoSuchKey error handling. - Updated documentation to reflect S3 dashboard storage migration. - Added tests for deploy_stacks.py and deploy.py — achieved 100% test coverage.	8	
		Redis/Valkey Row, MSK Fix, Datasources, Visual Polish & CLI Refactor	- Added Retro LCD bar gauges with duration units across all time panels for cleaner visual style. - Added renameByRegex to Performance Overview bar gauges for cleaner legend labels. - Applied consistent series colors across all timeseries panels for visual uniformity. - Fixed Row 6 MSK — replaced Bytes In metric with Messages Out per second for more meaningful throughput visibility. - Created monitored-resources.md documenting each AWS service monitored and metrics collected. - Built complete Row 9 Redis/Valkey with 11 panels covering the 4 golden signals — AMG 1-query compliant. - Renumbered all row files to sequential order row_01 through row_13. - Refactored CLI — replaced --dry-run/--validate-only with unified --validate flag and added --smoke-test option.	8	

Week 11	May 25 – May 29, 2026	Alarms Guide, Redis VPC, \$platform Refactor, SNS KMS & Alarm Panels	- Documented VPC datasource requirements and updated OpenSearch endpoint references. - Added Redis plugin placeholder panels noting VPC dependency for full connectivity. - Added redis-datasource to APPROVED_DATASOURCES and deployed provisioner v500. - Updated e2e and SQS tests to match current 13-row dashboard structure. - Refactored dashboard to use \${platform} variable in filterFieldsByName across all rows — removed hardcoded meridian. - Merged hotfix branch with platform variable refactor into release. - Performed alerts gap analysis documenting existing vs required alert coverage. - Created comprehensive alarms and notifications guide; updated notification email to bbhagat@citco.com.	8	42.5
		EventBridge + Lambda Alarm Panels & Decision Docs	- Documented alarm dashboard panel implementation options — EventBridge vs Scheduled Lambda. - Documented decision to proceed with Option A (EventBridge + Lambda) for real-time alarm panel updates. - Implemented EventBridge + Lambda integration for real-time CloudWatch alarm state changes in Grafana. - Improved alarm panels with color-coded state table and state-timeline history panel for visual alarm tracking. - Added table panel transformations and alarm dimension fields for richer data display.	4	
		ia_common_utility: Bitbucket Pipelines CI/CD Setup	- Added Bitbucket Pipelines CI/CD configuration and environment example file to ia_common_utility repo.	4	
		Alarm Panels, Scheduled Lambda, Grafana Alert Provisioning & LCD Barga	- Replaced state-timeline panel with retro LCD bar gauge for a more compact alarm state view. - Added scheduled Lambda with dynamic SQS alarms and panel improvements to the alerts row. - Added repeated alarm notifications using composite alarms and integrated Grafana alert panels. - Created alert rule provisioning script that pushes alert rules to Grafana via the Grafana API. - Made all alert panels full width (w:24) and stacked vertically for better readability. - Fixed Active CloudWatch Alarms panel to consistently show the latest alarm data. - Fixed Alarm State History bargauge to display all alarms with their latest state. - Removed filterFieldsByName from bargauge panel that was causing NO DATA.	8	
		Meetings & Standups	Meetings this week (10 meetings, 2.5h): - Daily Standup (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup (30 min) - Daily Standup (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Tech Talk Agentic Agents (30 min) - Daily Standup (30 min) - SRE Weekly Call (30 min) - Daily Standup (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min)	2.5	

Week 12	Jun 01 – Jun 05, 2026	RPA Production Support Shift	- Reviewed APAC handover log and opened the SD Portal queue for the start of Week 12 rotation. - Monitored UiPath Orchestrator for overnight queue items and job statuses across all active processes. - Triaged incoming tickets: reviewed MESO process completeness reports and confirmed successful overnight runs. - Handled FX Closeout status inquiry — verified Daily565 folder and confirmed report delivery to the correct location. - Checked VPL queue at 5:00 PM EST — all items processed, no manual retrigger required. - Updated Confluence shift handover log with end-of-day ticket summary and open items for APAC team. - Reviewed overnight ticket queue and APAC handoff notes. - SD Ticket #14951717 (GTL Not Generated — Voluntary Corporate Actions): Checked the Confluence page for Exception Handling — Corporate Actions, then the VCA section for the BPShare location. Triggered - SD Ticket #14952147 (AVCAB queue exception): Triggered the AVCAB queue process with -LOG suffix. Checked Blue Prism control panel — DailyVCAQueue showed exception. Reviewed VCAClientList and found err - Updated Confluence shift handover log with open items and escalation notes. - Reviewed APAC handover log and opened the SD Portal queue. - Auto Rec Download — CAPULA2 Config Update: Checked the CITCORECS_AUTO_REC_DOWNLOAD\SCRIPT UiPath share location. Created a backup of config.ini renamed to config-backup-2026-06-03.ini. Copied the file	40	40
		Alarm Bargauge Deduplication	- Fixed alarm state history deduplication in the Grafana bargauge panel by applying a merge transformation. - Simplified the bargauge panel by removing filter and merge transformations, switching to a clean GROUP BY AlarmName only query.		
		Alarm Panel	- Fixed duplicate alarms in the bargauge panel by correcting the GROUP BY clause and refining the regex pattern. - Added AlarmStateEncoded metric to support the LCD bargauge panel for visual alarm state representation. - Replaced the bargauge panel with a state-timeline panel for alarm history — provides a cleaner time-based view of alarm state changes.		
		CAIS Deadline SMBAuthenticationError Hotfix	- Fixed UCM credential validation by adding checks at startup and before SMB connections to catch SpnegoError authentication failures early — root cause of the CAIS Deadline filing failure. - Added a guard to prevent empty deadline_ids from generating an invalid SQL IN() clause. - Added a check to verify that a zip file exists before attempting to upload it, preventing silent upload failures. - Added full incident timeline and cascading impact analysis to the RCA documentation. - Created PR templates documenting all three hotfixes for the credential validation, empty deadline_ids, and missing zip upload guard. - Cleaned up temporary PR-TEMPLATE.md and PR-DESCRIPTION.md files from the repository. - Added unit tests for UCM credential validation, deadline_ids guard, and zip file upload guard. - Fixed email_function tests by mocking the constants module before import to avoid side effects.	8	

Week 13	Jun 08 – Jun 12, 2026	EC2 Monitoring Row & Instance Mapper Lambda	- Added new EC2 monitoring row with health and performance overview panels. - Corrected EC2 row position — placed before ECS Infrastructure for logical ordering. - Built EC2 instance mapper Lambda to support dashboard variable filtering by EC2 instance. - Added per-platform EC2 variables and wired them into panels using ec2_meridian. - Added dynamic ec2_instances variable sourced from a CloudWatch dimension query. - Removed WHERE clause from all EC2 panels to show all instances regardless of filter. - Moved EC2 CPU health panel to sit after Lambda and before ECS. - Bumped ProvisionerVersion after deploying EC2 monitoring changes.	8	42.5
		Health Row Split & CFN S3 Deploy Fix	- Refactored row_01_health by splitting it into individual per-panel files for better maintainability and modularity, consistent with the pattern used in other rows. - Fixed CloudFormation template deployment by uploading templates via S3 to work around the 51,200-byte inline template size limit.	8	
		Lambda/SQS SCHEMA SQL, Tooltip & Docs	- Updated e2e alarm tests to match the actual alarms.yaml file containing 18 alarms. - Fixed the Lambda row to use SCHEMA SQL syntax across all 37 meridian functions. - Added Row 1 Lambda Health panel explanation document and moved lambda inventory to docs/inventory/. - Added tooltip description to the Lambda Health panel; fixed tooltip to use \$platform variable. - Deployed Lambda SCHEMA SQL changes with tooltip and documentation updates (2 deploys). - Updated structure steering documentation to reflect new inventory/ and panels/ folders. - Fixed the SQS row to use SCHEMA SQL syntax and added Row 3 SQS panel explanations.	4	
		ia-it-sre-infrastructure-inventory Initial Scaffold	- Created new ia-it-sre-infrastructure-inventory repository — first commits for IISS-507. - Built initial Flask async API for resource inventory with async route handlers. - Aligned service_type names with dashboard row naming conventions and added BPM platform support. - Added Kiro steering files, project documentation, and pre-commit configuration.	4	
		SQS Monitoring Dashboards & Platform Filter	- Added SQS monitoring dashboards to ia-resources-monitoring with platform filter support across all resource types, extending CloudWatch dashboard coverage to SQS queues.	8	
		Meetings & Standups	Meetings this week (8 meetings, 2.5h): - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - Tech Talk Idea Box Agent (part of the Solution...) (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - Citco Works Monitoring Requirements (60 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min)	2.5	

Week 14	Jun 15 – Jun 19, 2026	Resource Tables, Row Layout Restructure & Drill-Down Links	- Added missing panel documentation for rows 7-10 and numbered inventory files by row for easier navigation. - Added data links from Row 1 health panels to Row 2 performance panels for drill-down navigation. - Added renameByRegex to 9 resource table panels and updated related documentation. - Added resource table panels with clickable AWS Console links to all service rows for quick incident navigation. - Restructured the SQS row with separate Queues and DLQ tables and a 4-panel metric row at w=6. - Fixed Row 1 SQS Health stat panel to exclude DLQ queues from the count. - Enhanced performance SQS panel to exclude DLQs and show only top K queues; added SQS Health description. - Removed sortBy from SQS tables and used max reducer for the DLQ table.	4	42.5
		Inventory API Coroutine Fix & All Discoverers	- Fixed coroutine serialization issue in the inventory API. - Implemented all resource discoverers. - Added profile and metrics support to the inventory API.	4	
		Deep-Links, Legend Fixes, Rows 10-13 Restructure & Performance Row	- Fixed all ECS APIs panels to display clean target group names in legends. - Fixed request throughput, HTTP 4xx/5xx, and p50/p95/p99 panels to show clean ALB and target group names. - Fixed MSK resource table link to use cluster search filter for accurate console navigation. - Merged feature/IISS-487-link-panels into release. - Documented AWS Console deep-link limitations and available workarounds; added Infinity datasource plugin guide. - Added bargauge deep-links to ECS Top CPU panels — each bar links directly to the ECS Console service page. - Added SQS bargauge deep-links for Top Queues by Depth and Message Age panels linking to SQS Console. - Added EC2 panels with deep-links and moved EC2 row before ECS; documented EC2 memory limitations.	8	
		Redis Restructure, OpenSearch Status Tables, Health Row Deep-Links & P	- Reorganised Row 11 Redis into 12 panels across 3 visual rows; replaced Redis CPU bargauge in Performance row with Memory Usage and Cache Hit Rate using inverted thresholds for worst-first display — ProvisionerVersion 698 - Updated panel documentation for Redis row and performance row including AVG vs MAX rationale. - Added 3-way merge strategy rule to workflow steering docs. - Replaced single OpenSearch cluster health stat with 4 separate status tables (Green, Yellow, Red, Not Available) using two-step reduce pattern to handle CloudWatch float/integer mismatches — ProvisionerVersion 706. - Documented OpenSearch status table design decisions and reduce transformation pitfalls in known-limitations.md. - Restructured Row 13 OpenSearch into 12 focused SRE panels across 3 rows (status, capacity, reliability). - Added OpenSearch JVM Memory Pressure bargauge to Performance row with 75%/90% thresholds — ProvisionerVersion 712. - Fixed RDS health panel to collapse multiple DB instances into a single MAX value via two-step reduce transformation.	8	

		Lambda/SQS/EC2 Variable Filters, Tested & Reverted Approaches	- Added lambda_function and sqs_queue dashboard variables for cross-panel filtering. - Documented limitations of table filters vs dashboard variables for future reference. - EC2 instance variable filtering added but deferred — variable returned empty results. - Tested and reverted CloudWatch label override experiment — confirmed broken with Metric Insights SQL mode; documented as known limitation.	2	
		ECS Service & Target Group Variables	- Added ecs_service and ecs_target_group variables for cross-panel filtering across ECS rows. - Fixed ECS Infra legend format to match resource table and applied anchored regex filter. - Fixed ECS APIs target group names in dropdown and cleaned p50/p95/p99 legends.	2	
		MSK Consumer Group Variable	- Added msk_consumer_group variable for filtering consumer lag data in the MSK row — key SLI from the April 17 Meridian Incident.	2	
		DB Instance Variables (DocumentDB, RDS, Redis, Neptune, OpenSearch)	- Added docdb_instance, rds_instance (with Aurora regex to exclude DocumentDB), redis_cluster, neptune_instance, and opensearch_domain dashboard variables — completing row-level filtering across all database service rows. - rds_instance variable uses Aurora regex pattern to correctly exclude DocumentDB instances from the RDS dropdown.	2	
		Performance Row SCHEMA SQL Migration, Freeable Memory Panels & EC2 Var	- Migrated 6 legacy SQL panels to SCHEMA syntax; reduced all performance row bar gauges from top-10 to top-5 with names displayed above bars. - Fixed missing end-of-file newlines in 19 JSON panel files and applied ruff formatting cleanup. - Added Neptune Freeable Memory panel to the Performance row; aligned thresholds across all DB freeable memory panels. - Set fixed 16 GiB scale on all freeable memory bar gauge panels for visual comparability across DB services. - Corrected freeable memory thresholds to use actual per-instance-class memory sizes (verified via describe-db-instances) with percentage-based threshold mode. - Merged Performance row improvements and freeable memory panels into release. - Fixed EC2 variable namespace mismatch, reordered variable position to match dashboard row order, and removed allValue wildcard so "All" interpolates as actual instance IDs. - Removed unused LambdaCodeBucket/Key CloudFormation parameters and renamed the ECS panel.	8	
		Meetings & Standups	Meetings this week (10 meetings, 2.5h): - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - BPM Flight Control - Platform & Dev (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - SRE Weekly Call (30 min) - Daily Standup - Innovation IT (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Instrument Expiry - Support Exception Handling (30 min)	2.5	
		EC2 Variable Filter & Mapper Deploy	- Added EC2 variable filtering to EC2 health, performance, and resource table panels — shows only meridian-tagged instances. - Merged feature/IISS-487-link-panels into release, completing the deep-link and variable filtering feature set.	4	

		SQS Queue Mapper Lambda & Panel Filtering	- Fixed SQS panel filters to use case-insensitive regex — resolved issue where ia-bpo platform showed only 2 of 55 queues due to uppercase/lowercase mismatch. - Researched Infinity plugin as SQS integration approach — documented planning, connectivity options, and postmortem (abandoned due to unresolvable Go HTTP client timeout issue). - Built SQS Queue Mapper Lambda discovering queues by name patterns and tags, publishing custom metrics to all 6 platforms — resolved 36 previously invisible queues. - Refactored Lambda package with shared platforms.py module used by all mapper Lambdas. - Replaced 5-minute schedule trigger with CloudTrail event-driven triggers — reduced Lambda invocations from ~8,640 to ~2-5 per month. - Deployed SQS Queue Mapper Lambda — verified 139 queues across all 6 platforms. - Updated SQS panels to filter by sqs_queue variable and added AWS Console data links to all timeseries panels. - Moved EC2 and SQS mapper Lambdas to dedicated resource-mappers.yaml CFN stack.	4	
		Lambda/MSK Mapper Lambdas, Idle State, Variable Filtering & MSK Deep-L	- Fixed health panels to show 'Idle' instead of 'No data' when no metrics are available; added noValue='Idle' to all SQS row panels for consistent empty-state display. - Built Lambda function mapper Lambda — discovers Lambda functions and updates the dashboard variable and panels accordingly. - Fixed SQS performance panels to use sqs_queue:pipe variable format for correct multi-value filtering. - Updated row-04-lambda documentation to reflect current panel implementation. - Added docs_generator package for automated panel documentation generation. - Added invoke_mapper and inspect_s3_dashboard utility scripts. - Built MSK cluster mapper Lambda with console deep-links on all MSK panels. - Removed platform filter from MSK panels and fixed ARN-based deep-links using percent-encoding.	4	
		Inventory API Standardisation & Bulk Tag Discovery	- Standardised inventory API resource response format and added a CLI entry point. - Migrated all discoverers to the Resource Groups Tagging API for bulk tag discovery. - Fixed per-resource tag checks with throttle handling to match the mapper Lambda pattern.	4	
		MSK Deep-Links Complete, Mapper Schedule, Consumer Lag Split & Deploy	- Iterated on MSK cluster mapper schedule — tested 6-hour, 5-minute, and finally CloudTrail event-driven triggers for CreateCluster/DeleteCluster events; made schedule rate configurable via CFN parameter. - Added comprehensive resource mappers documentation covering all mapper Lambdas, deploy order, and known limitations. - Completed full rewrite of MSK documentation covering panels, variables, deep-links, and known limitations. - Integrated resource-mappers CFN stack into automated deploy pipeline — established deploy order: resource-mappers → grafana → alarms. - Added ASCII and Mermaid diagrams explaining MSK cluster/topic/consumer group hierarchy. - Updated MSK inventory to include dev-mas-observability-cluster and reclassify axon-msk-cluster as tag-matched. - Added ListMetrics RecentlyActive verification guide, then corrected it after source-confirming Grafana v12 does not use the RecentlyActive parameter. - Implemented Option B for MSK consumer lag deep-links — added ClusterArn as a third dimension to MSK mapper metrics, enabling chained variable for ARN resolution.	8	

Week 15	Jun 22 – Jun 26, 2026	6 Resource Mapper Lambdas, EC2 Row Restructure & Deploy Script Fixes	- Added Broker CPU Max panel to MSK row — broker row now 4 panels at w=6 for consistent layout. - Created planning document for expanding resource mappers to ECS, DocDB, RDS, Redis, Neptune, and OpenSearch. - Added EC2 instance mapper to resource-mappers CFN stack — was previously implemented in code but never deployed, causing ec2_instances variable to always return empty. - Fixed duplicate reduce transformation in EC2 health panel and aligned mapper with shared PLATFORMS pattern. - Rewrote invoke_mapper.py with multi-resource invocation and --all flag for running all mappers at once. - Added exit code taxonomy documentation and per-run log files to logs/ for deploy and invoke-mapper scripts. - Fixed EC2 panel transformation order — filterFieldsByName must run before renameByRegex across all 8 panels. - Fixed deploy script: proper exit codes, command logging, and UPDATE_IN_PROGRESS state detection.	4	42.5
		Concurrent Tag Discovery	- Added concurrent tag discovery with 10 worker threads (client-per-thread pattern) to the inventory API for significantly improved performance on large fleets.	4	
		Lambda Deploy Fix, Row Structure Refactor, Health/Performance Overview	- Fixed critical Lambda code deployment gap — introduced SHA256 content-hash S3 keys so CloudFormation detects actual code changes instead of silently skipping unchanged Lambdas. - Removed redis-datasource plugin references from dashboard — VPC connectivity no longer needed. - Reordered dashboard rows and renamed folders to full service names for better clarity. - Restructured Row 1 Health Overview into 2 visual sub-rows; added ECR/EKS placeholder panels. - Restructured Row 2 Performance Overview into 6 visual sub-rows; added ECR/EKS/ECS-5xx panels. - Fixed build script to generate stable panel IDs; corrected Row 1 health panel deep-links. - Fixed OpenSearch health panel links to correctly point to Clusters Red/Yellow panels. - Merged row structure refactor into release and reordered dashboard variables to match row order.	4	
		127 Unit Tests, CLI Entry Point, State Tracking & Docs	- Renumbered inventory discoverers and renamed service types to align with Grafana dashboard row naming. - Added CLI entry point with --env flag and automatic corporate certificate check on startup. - Added 127 comprehensive unit tests covering all inventory modules. - Added comprehensive documentation with architecture diagrams, API reference, configuration guide, and contributing guide. - Added .env.example, removed __future__ imports, and made paths platform-agnostic. - Clarified tag-key-agnostic filtering and added two-pass flow diagrams to documentation. - Added state/status tracking to MSK and OpenSearch discoverers for richer resource information.	4	

		Meetings & Standups	Meetings this week (11 meetings, 2.5h): - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - Support KT - FX Closeouts Process Exception (60 min) - Tech Talk AURA (part of the Solution Snapshots) (30 min) - Support KT - FX Settled Process Exception Handling (60 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - SRE Weekly Call (30 min) - Daily Standup - Innovation IT (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min)	2.5	
Week 16	Jun 29 – Jul 03, 2026	RPA Production Support Shift	- Reviewed APAC handover log and opened the SD Portal queue for the start of Week 16 rotation. - SD Ticket #15103842 — Corporate Actions new client onboarding (Leeside): Checked Confluence page for Corporate Actions Processes — confirmed process migrated to UiPath. Checked UiPath Cloud CoE/HedgeF - SD Ticket #15102495 — Add Equity Fund Product Funds to OTC Process for Lumyna: Checked Confluence pages for OTC Confirmations documentation, exception handling guide, and solution design document — no - Monitored VPL queue at 5:00 PM EST — all items processed, no manual retrigger required. - Updated Confluence shift handover log with open items and next-day follow-ups. - Reviewed APAC handover log and SD Portal queue; followed up on SD #15102495 (OTC Lumyna) — awaiting response from Miko/Daniel on MNL team handover. - SD Ticket #15113393 — Corporate Actions new client onboarding (ORCHID_TREE): Checked Confluence page for Corporate Actions Processes — confirmed UiPath migration. Checked UiPath Cloud CoE/HedgeFunds/V - SD Ticket #15111049 (PreMeso Config — SharePoint): Reviewed the ticket and checked the PreMeso SharePoint configuration settings. Investigated the relevant Confluence page for PreMeso config parameter - Monitored MESO queue throughout the shift — all runs completing within SLA. - Checked VPL queue at 5:00 PM EST — queue clear, no retrigger required. - Reviewed and updated Confluence shift handover page: documented all open ticket statuses, escalation notes for OTC Lumyna (MNL team dependency), and flagged PreMeso Config as pending end-user response - Reviewed APAC handover log; lighter queue expected due to Canada Day public holiday.	40	40

		Health Overview Optimisation, MSK Consumer Lag & Mapper Rename	- Refactored mapper Lambda folders with row number prefixes for consistent naming alignment with the dashboard. - Removed MonitoringBucket from CloudFormation (now managed externally) and fixed mapper-to-parameter name mappings. - Optimised health stat panels — removed seriesToRows transform, added ORDER BY MAX() DESC for correct value display (required revert-and-redo to land cleanly). - Added variable-based filtering to health panels and introduced new ecs_cluster dashboard variable. - Fixed MSK topic deep-link URL format to match correct AWS Console path; rebuilt assembled dashboard. - Updated MSK consumer lag by consumer group panels to show ClusterName/ConsumerGroup in legends. - Added msk_cluster variable filtering and cluster console links to consumer lag and estimated max time lag panels. - Added consumer lag by topic panel with GROUP BY Cluster Name + Topic and cluster/topic legends.		
		MSK Health Stat Alignment	- Improved MSK health stat panel to GROUP BY Cluster Name + Consumer Group with msk_cluster:pipe filtering. - Aligned MSK performance panel to use the same GROUP BY and filter for consistency.	2	
		ECR Monitoring Row	- Added complete ECR monitoring row — mapper Lambda, resource table panels, dashboard variables, and deep-links. - Added X-Ray tracing implementation estimate document.	2	
		EKS Monitoring Row	- Added complete EKS monitoring row — mapper Lambda, real metric panels, dashboard variable, and deep-links. - Removed unused msk_consumer_group variable and updated related documentation. - Fixed MSK performance panel link to use correct AWS Console URL format.	2	
		S3 Cache Layer, CI/CD Pipeline, Naming Convention & SSL Fix	- Merged S3 cache layer, CI/CD pipeline, and deployment scripts into release. - Added CI/CD pipeline for citcodevops01 account with full deployment guide. - Renamed all AWS resources to ia-it-sre-resources-inventory- naming convention. - Added profile parameter as dropdown in Swagger UI with named options — bumped API version to 1.1.0. - Fixed SSL crash in S3 cache when running on VPN — now degrades gracefully with helpful hint (v1.1.1). - Added certifi as explicit dependency to fix missing package error on Python 3.14 (v1.1.2). - Added AWS_CA_BUNDLE to .env examples with setup instructions for corporate VPN environments. - Replaced hardcoded account IDs with placeholders across all .env example files.	2	
		Panel Link Fix, Semantic Versioning, X-Ray Spec & Inventory Comparison	- Fixed health panel links to panel-level and added . wildcards to all :pipe variable filters for correct multi-value interpolation. - Added semantic versioning steering rules to the project documentation. - Added X-Ray tracing specification covering requirements, design, and implementation tasks. - Added platform-agnostic rename migration plan and finalized naming convention spec. - Added X-Ray infrastructure requirements and progress tracking document. - Renamed inventory documentation folder to meridian-service-inventory and added resources-inventory comparison folder. - Completed mapper vs inventory repo comparison document.	8	

Week 17	Jul 06 – Jul 10, 2026	CloudWatch Metrics Publishing, Pydantic Models & S3 Key Refactor	- Added CloudWatch metrics publishing from inventory cache Lambda — deployed 597 metrics across 6 platforms to DEV. - Fixed MSK to publish cluster names (not topics) and Redis to publish cache node IDs (not replication group IDs) to match what Grafana dashboard variables expect. - Added has_metrics and emitting_metrics fields to S3 cache so cached responses are identical to live API responses. - Fixed API refresh path to write has_metrics to S3 before returning the response. - Added Pydantic response models with 12 service-specific resource types and S3 versioned key support (latest.json + history). - Added Grafana integration guide documenting the resource discovery → S3 → CloudWatch → Grafana variable flow. - Fixed Lambda dependency packaging — added missing typing_extensions and annotated_types, used Linux platform target for compiled extensions. - Refactored S3 keys to use {platform}/{row_name}/ structure for easier browsing.	4	42.5
		Platform-Agnostic Rename (ia-it-sre Convention) & Docs Reorganisation	- Updated rename spec with full migration plan covering Phases 8-18. - Renamed all CloudFormation stack names, Grafana execution role, workspace IAM roles, Lambda functions, dashboard UID, alarm descriptions, and pipeline stack names from meridian to the ia-it-sre naming convention. - Replaced remaining meridian references throughout the codebase. - Reorganised docs/ into guides/, operations/, services/, limitations/, and decisions/ subfolders. - Updated all documentation to reflect the ia-it-sre naming. - Fixed e2e and CloudFormation tests to pass with the new ia-it-sre naming.	4	
		Infinity Datasource Migration, Platform Rename Complete, EC2 Name Tag	- Completed overnight test/refactor: replaced stale row folder tests with dynamic validation, fixed docs_generator row numbering, and split generators/ into rows/ and inventory/ subpackages. - Fixed preflight to fall back to workspace API check when CFN stack not found — handles first-deploy scenario. - Merged platform-agnostic rename refactor into release. - Installed Infinity plugin on AMG and created S3-Resources-Inventory datasource with SigV4 auth. - Attempted direct S3 variable migration via Infinity, reverted due to SigV4 issues, then successfully migrated all 14 dashboard variables to Infinity via CloudFront (CloudFront handles S3 signing via OAC). - Fixed pipe variable filterFieldsByName regex grouping across 19 panels for correct multi-platform filtering. - Renamed dashboard title to IA IT SRE Resources Monitoring. - Completed platform-agnostic rename across all remaining phases — workspace template, IAM roles, provisioner policy, pipeline stacks, alarms, docs, tests, and remaining code references.	4	

		CloudFront S3 Proxy, DB Discoverer Enhancements & Package Refactor (v1	- Added Infinity datasource + mapper decommission spec and CloudFront S3 proxy spec for Grafana variables. - Added CloudFront + S3 OAC proxy CFN template for serving inventory JSON to Grafana without client-side SigV4. - Enhanced RDS and DocumentDB discoverers to return both clusters and instances (v1.3.1). - Added tab=monitoring to DocumentDB console URLs for direct monitoring view. - Added Redis/Valkey clusters and instances with engine differentiation; fixed console URL paths and member cluster name matching (v1.3.2). - Added Neptune clusters and instances with engine, type, and console URLs (v1.3.3/v1.3.4). - Refactored inventory into sub-packages: core, api, infra, and metrics — bumped to v1.4.0. - Fixed EKS URL to use console link; fixed ECR URL to use console repository link.	4	
		Hotfix Merge, Health Panel Fixes & Named Exit Codes	- Merged hotfix with variable-based panel filters, alarm fixes, and fresh Grafana workspace provisioning. - Added full panel tables for all remaining rows (EC2, ECR, EKS, RDS Aurora, DocumentDB, Neptune) to the Confluence guide. - Refactored health panels to move LIMIT 20 from a transformation step into the SQL query directly. - Added renameByRegex to all 14 health overview panels to strip CloudWatch ORDER BY numeric prefix from series names. - Added named exit code constants to main.py CLI for clearer error handling.	2.75	
		X-Ray Tracing Row & Documentation	- Built new X-Ray tracing Row 17 with 4 panels: Service Map, Recent Traces, Response Time p50/p95/p99, and Error/Fault Rate using the X-Ray datasource; added xray: read permissions to the Grafana workspace IAM role. - Added X-Ray self-service enablement guide with Lambda tracing commands, ALB setup, and cost estimates. - Added team request documentation for ECS sidecar tracing (IISS-839), EKS ADOT (IISS-840), MSK instrumentation (IISS-843), API Gateway (IISS-841), and database client instrumentation (IISS-844). - Added X-Ray limitations document covering AMG 12.4 constraints plus Application Signals overview for future SLO-based monitoring.	2.75	
		Confluence Docs, S3 Lifecycle Rules & v1.4.3	- Synced docs with v1.4.1 covering instances-only pattern and EKS/ECR console URLs. - Added 5 Confluence documentation pages: executive overview, architecture, API user guide, developer onboarding, and deployment runbook (v1.4.2). - Updated .gitignore to exclude PR docs and Confluence docs from tracking. - Added 7-day S3 lifecycle expiration rules for history/ objects with 78 prefix-based rules across all platforms and services (v1.4.3).	2.5	

		Meetings & Standups	Meetings this week (11 meetings, 2.5h): - IA Config Portal - DEMO (30 min) - BPM 3.25 Handover (6 JIRAs) (30 min) - Daily Standup - Innovation IT (30 min) - BPM Flight Control - Platform & Dev (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - IA CONFIG PORTAL - DEMO (45 min) - Blue Prism Troubleshooting (60 min) - BPM Flight Control - close out actions for 3.25 (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min)	2.5	
		EC2 Deep-Links & IISS-485: EKS Deep-Links	- Added AWS Console deep links to EC2 and EKS time series panels. - Removed duplicate filterFieldsByName transformations from DB time series panels.	1	
		ECS Deep-Links, Mapper Decommission & Infinity Migration	- Added AWS Console deep links to ECS service and ALB Target Group time series panels. - Decommissioned all mapper Lambda handlers; migrated ECR and ECS resource tables to Infinity datasource. - Fixed ecs_target_group variable to remove allValue wildcard that was bypassing platform filtering.	1	
		DB Time Series Deep-Links & Variable Simplification	- Added AWS Console deep links to RDS, DocumentDB, Redis/Valkey, and Neptune time series panels. - Simplified DB dashboard variables from 8 (cluster+instance pairs) down to 4 single variables per service. - Applied simplified DB variables to assembled dashboard JSON.	1	
		Alarms, Grafana User Management, Custom URL, Agents Extraction & Platf	- Updated Project tag to IA-IT-SRE-Resources-Monitoring; added ECS CPU critical alarm at 80%, lowered warning to 60%. - Enhanced Grafana user management CLI with list/add/update/revoke commands and user resolution; documented DL/SSO group limitations and bulk access guidance. - Attempted Route53 CNAME for custom dashboard URL — removed after hitting 403 on private zone. - Added CloudFront reverse proxy as the working solution for a custom Grafana dashboard URL; documented AMG Host header limitation. - Moved logging config out of agents/ into scripts/log_config in preparation for extraction. - Extracted agents/ package into new standalone repo ia-it-sre-agent-orchestration. - Renamed all references from ia-meridian-monitoring to ia-it-sre-resources-monitoring-amg across docs and configs. - Reordered ECR table columns — RepositoryName first, Images second.	1	
		MSK Inventory-Driven Deep-Links	- Migrated MSK row to inventory-driven deep-links using the Infinity datasource for all MSK panels.	1	
		ia-it-sre-agent-orchestration Initial Commit	- Created new ia-it-sre-agent-orchestration repository — agents package extracted from the monitoring repo for standalone reuse. - Added agentic AI guide from the monitoring repo.	1	
		MSK Topics Filter	- Filtered internal Kafka topics (__ prefix) from the inventory API. - Scoped topics to their platform's name patterns for cleaner results (v1.4.4).	1	

Week 18	Jul 13 – Jul 17, 2026	Dynamic Alarm Provisioner, S3-Backed Alarm Lambdas & Notification Form	- Built full S3-backed alarm Lambda architecture: alarm_notifier, alarm_re_notifier, and alarm_state_publisher with shared models/formatter layer. - Implemented dynamic alarm provisioner Lambda — reads resource inventory from S3 and automatically creates/updates per-resource CloudWatch alarms for all 6 platforms. - Replaced 17 static CloudWatch alarms with the dynamic provisioner covering ECS, RDS, DocumentDB, Redis, Neptune, and more. - Added 9 missing alarm types: Lambda error rate (math expression), Lambda throttles, SQS message age, MSK consumer lag, MSK broker disk, EC2 status check, ECS running vs desired tasks, OpenSearch cluster health, and Docum - Standardised all CPU thresholds to 60% warning / 80% critical; adjusted SQS depth to 5/10 and MSK lag to 1000/5000. - Added pytest tests for all alarm Lambdas and shared modules. - Fixed concurrent alarm creation — reduced ThreadPoolExecutor to 2 workers with adaptive retry (up to 10 attempts) to avoid PutMetricAlarm API throttling; increased timeout to 900s. - Iterated on notification formatting: emoji-based → ASCII table → clean key-value fixed-width label format.	2.75	42.5
		SQS Panel Updates	- Updated SQS health and queues table panels.	2.75	
		ALB Target Group Discoverer	- Added ALB target group discoverer to the inventory API — enables dashboard variable migration from ECS mapper Lambda to Infinity datasource for ecs_target_group filtering.	2.75	
		Alarm Panel Fixes, Lambda Rename & Alarm Definition Refactor	- Fixed alarm state history panel height, enabled scroll for overflow, and filtered to show only fired alarms for the selected platform with the correct IA-IT-SRE regex pattern. - Reverted unsupported SQL WHERE clause on metric values — landed on ORDER BY MAX() DESC LIMIT 50 as the working approach. - Merged alarm panel hotfixes into release. - Renamed all alarm Lambdas to lda-ia-it-sre- naming convention. - Rebuilt assembled dashboard JSON (33 panels). - Extracted alarm definitions into per-service modules (rows 03-15) with a centralised thresholds.yaml config file. - Added new alarms for ECS memory, EKS CPU/memory, and split EC2 into instance check and system check alarms. - Fixed alarm name assertions in tests to match new row-based ID prefix format.	4	
		X-Ray Groups, SDK Instrumentation & Grafana Datasource	- Created X-Ray tracing groups for all 6 platforms (meridian, citcoworks, automationhub, ia-bpo, bpm, rpacfs1) and enabled Active tracing on all 4 alarm Lambdas. - Documented X-Ray enablement status — groups deployed; Meridian platform tracing pending team action. - Added aws-xray-sdk instrumentation to all 4 alarm Lambda handlers using patch_all() for automatic boto3 subsegment tracing. - Updated packaging script to bundle aws-xray-sdk and dependencies into Lambda zips (~9KB → ~170KB each). - Added SDK instrumentation request as a critical blocker for platform teams in the X-Ray documentation. - Added X-Ray datasource and read-only IAM permissions to the Grafana workspace via IaC. - Replaced trace volume timeseries panel with Recent Traces table showing trace IDs with clickable links to Grafana Explore. - Rebuilt assembled dashboard JSON (v62, 33 panels) and merged X-Ray feature into release.	4	

		Alarm Panel Fix & Row 10 Alarm Docs	- Documented Row 10 alarm requirements and their dependency on the inventory repo's Phase 4 target group data. - Fixed alarm name regex in the dashboard alarms panel to match the new IA-IT-SRE prefix; switched period to auto for better resolution. - Merged alarms-notifications feature into release.	2.75	
		X-Ray Row Completion, Application Signals Phase 1 & Provisioner Auto-C	- Completed X-Ray dashboard row with 4 panels: Service Map (nodeGraph), Trace Volume Over Time (stacked bars by group), Traces by Group (horizontal bar gauge), and Recent Traces table with clickable trace IDs. - Fixed panels 5/6 to use CloudWatch AWS/X-Ray metrics — X-Ray plugin does not support getTimeSeriesServiceStatistics. - Marked xray-tracing spec as complete; updated service map tooltip with datasource error workaround. - Added Application Signals read permissions (7 IAM actions) to the Grafana workspace role and enabled account-wide discovery. - Completed Application Signals Phase 1 — IAM setup, discovery enabled, dashboard link to Application Signals AWS Console. - Added auto-creation of X-Ray datasource in the provisioner Lambda on every deploy. - Renamed Grafana workspace from ia-it-sre-resources-monitoring to ia-it-sre-infrastructure-monitoring; updated all references.	2.75	
		ECS Target Group Discoverer	- Increased inventory API test coverage from 62% to 99% with 453 tests including Hypothesis property-based tests (v1.4.6). - Added target_group_full_name and alb_full_name fields to ECS target group discoverer with ARN-based console URLs (v1.4.7). - Renamed row_10_ecs_apis S3 cache path to row_10_ecs_target_groups; documented breaking change migration tasks in Phase 5 spec.	2.75	
		X-Ray SDK Fix, App Signals Phase 2, Docs Consolidation & Observability	- Renamed Grafana dashboard title to IA IT SRE Infrastructure Monitoring; moved S3 bucket rename to a separate spec. - Fixed X-Ray tracing by adding explicit service name configuration via xray_recorder.configure(service=...) to all 4 Lambda handlers — without this, traces exist but are not discoverable by X-Ray groups. - Documented the xray_recorder.configure(service=...) requirement for future Lambda instrumentation. - Consolidated 8 X-Ray documentation files into 5 focused documents (team-requests, progress, limitations, gap-analysis, cost-analysis) — eliminated all overlap. - Marked Application Signals Phase 2 complete (datasource auto-provisioning). - Rewrote gap-analysis and cost-analysis documents with a full capability matrix showing parity progression: CloudWatch alone 30% → full stack 88%. - Added Dynatrace feature descriptions and pairwise tool comparison. - Added comparison showing CloudWatch + X-Ray + App Signals achieves 78% parity without Grafana.	4	
		S3 Cache Region-Aware Prefix Fix	- Fixed inventory S3 cache prefix to be region-aware — ensures correct path structure across multi-region deployments.	4.5	

		Meetings & Standups	Meetings this week (10 meetings, 2.5h): - CORA Support Handoff <> Final Connect (60 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - RPA Deployment Pause - Next Steps (60 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - SRE Weekly Call (30 min) - Daily Standup - Innovation IT (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min)	2.5	
		X-Ray Datasource Fixes, nodeGraph Workaround & Trace Backends Comparis	- Restored X-Ray nodeGraph service map panel and removed duplicate traced services panel. - Added trace backends comparison doc: X-Ray vs Tempo vs Jaeger vs Zipkin with feature matrix, cost analysis, and architecture options. - Replaced broken nodeGraph panel with AWS Console link panel — plugin crashes for viewer-role users (unfixable at dashboard level). - Fixed X-Ray datasource UID to stable provisioner-managed value xray-ia-it-sre-dev across all panels. - Fixed CloudWatch provisioner to use stable 'cloudwatch' UID matching all 100+ panel references; removed unnecessary OpenSearch datasource creation.	1.25	
		ECS Target Group Alarms & Row Rename	- Implemented ECS target group alarms (rows 10.0-10.3). - Renamed row from ecs_apis to ecs_target_groups for consistency with inventory naming.	1.25	
		Powertools Migration, VPN SSL Hotfixes & Cross-Platform Data Leak Fix	- Migrated all alarm Lambdas from aws-xray-sdk to aws-lambda-powertools for tracing. - Added AppName and BudgetCode (IA-BPO) tags to all Lambda functions for cost attribution. - Removed empty Grafana Alarms panels from row 16. - Resolved VPN SSL hotfix series: fixed AWS_CA_BUNDLE path resolution, added certifi.where() fallback, bundled aws-lambda-powertools directly in Lambda zips, moved cfnlint import inside try block, added colorama for Window - Added VPN SSL deploy fix documentation. - Fixed cross-platform data leak: empty variable values caused filterFieldsByName to match all platforms — added allValue='^\$' __no_resources__ sentinel across all 13 multi-select variables.	1.25	
		ECS Target Group Discovery Phase 4 Complete	- Completed ECS Target Group Discovery Phase 4 — ARN-based console URLs, target_group_full_name and alb_full_name fields (v1.4.8). - Fixed S3 cache mocking in integration tests to prevent real S3 writes. - Added AppName and BudgetCode (IA-BPO) tags to Lambda functions.	1.25	
		X-Ray Powertools Instrumentation & Lambda Trigger Classification	- Added X-Ray tracing via AWS Lambda Powertools to inventory service with platform/service/environment annotations (v1.4.9). - Added trigger_type classification field and spec to Lambda resource model.	1.25	
		Lambda API Migration v1.5.0	- Deprecated Flask API and migrated to 4 dedicated Lambda handlers behind API Gateway (v1.5.0). - Added CloudFormation template and deploy-api CLI command for the new Lambda API. - Added /docs endpoint with embedded Swagger UI and profile dropdown. - Implemented async background refresh to avoid 30-second API Gateway timeout. - Added /refresh endpoint supporting per-platform, per-service, and full refresh targeting.	1.25	

Week 19	Jul 20 – Jul 24, 2026	Variable Fix, VPN Cert Deploy & noValue Fallback	- Fixed health row panels to remove trace links and corrected allValue from __no_resources__ to . for all multi-select variables — resolves panels showing Idle/no data when resources exist. - Fixed deploy script to auto-patch venv certificates with the corporate CA bundle and switched Lambda packaging to use venv pip — resolves VPN SSL failures during deploy. - Added noValue fallback text to all Infinity resource tables for graceful empty-state handling.	4	42.5
		Resource Tables to Infinity Datasource, Lambda Tags & Pipeline Auto-De	- Converted all resource tables across the dashboard from CloudWatch-based queries to the Infinity datasource pulling from the S3/CloudFront inventory API — tables now show all discovered resources regardless of CloudWatch - Standardised Lambda tags with AppName=ia-it-sre-resources-inventory and BudgetCode=ia-bpo. - Reorganised inventory documentation into grouped subdirectories: api/, architecture/, deployment/, development/, confluence/. - Updated Bitbucket pipeline to deploy the Lambda API stack automatically and invoke the refresh Lambda post-deploy to pre-populate the S3 cache.	4	
		EKS API 5xx Alarm	- Added EKS API 5xx alarm and synced threshold to the performance panel.	2.75	
		Alarm Panel Fixes, Platform Config, Alert Health Panels, Limitations D	- Added clickable AWS Console links to alarm names in the Active CW Alarms panel using renameByRegex to strip numeric prefix. - Fixed alarm panel links and regex patterns to correctly reconstruct full alarm names for console deep-linking. - Set alarm state panel period to 300s to match the publisher interval. - Changed alarm metric namespace from per-platform to unified ia-it-sre/Alarms covering all 6 platforms. - Centralised platform list in config/base.yaml as single source of truth; deploy script injects it into CloudFormation parameters automatically. - Removed legacy unused alarm parameters and retired resource-mapper parameter files. - Created standalone GLOSSARY.md and comprehensive cost-analysis and gap-analysis docs for Confluence. - Iterated on health alert panels — fixed platform filtering, alarm severity patterns, label overrides, and reduce transformations for correct firing alarm counts.	2.75	
		Naming Standardisation, Custom Domain & Test Coverage Restoration	- Fixed property tests to mock _cache preventing real S3 writes during tests. - Removed hardcoded API URLs and standardised all naming from ia-sre to ia-it-sre across the repo. - Added custom domain support: ia-it-sre-resources-inventory-{env}.citco.com via API Gateway V2 domain mapping. - Updated all documentation with consistent custom domain URLs and Swagger UI paths. - Added GLOSSARY.md and AWS Lambda Powertools section to contributing guide. - Bumped version to 1.5.1. - Added 50 unit tests for Lambda API handlers and 48 additional tests — restored coverage to 99% with 562 total tests.	2.75	

		Composite Alarms, Notifier Fixes & Auto Invoke Provisioner	- Fixed alarm notifier — removed duplicate tracer/import causing WARNING notifications to fail silently. - Fixed alarm formatter to remove stale datapoints from ongoing re-notification emails. - Fixed Lambda packaging to include pyyaml dependency and .yaml file for the provisioner. - Added post-deploy auto-invocation of the alarm provisioner (async, non-blocking) for hands-free deployment. - Added 6 same-resource composite alarms for noise reduction across DocumentDB, RDS, Redis, OpenSearch, and ECS. - Added composite alarm suppression in the re-notifier — skips child re-notifications when a composite alarm fires. - Added branch protection steering rules (release=merge-only, feature=merge-release-first). - Rewrote alarms-and-notifications operations documentation with full architecture and provisioner system details.	4	
		Spec Status Update & infrastructure-inventory Rename	- Updated all spec task files to reflect actual completion status. - Renamed inventory project from resources-inventory to infrastructure-inventory across Lambda names, CloudFormation stacks, documentation, and URLs.	4	
		X-Ray Trace ID Single-Click Fix & Comprehensive Team Requests	- Fixed X-Ray Recent Traces table to use a single plugin-provided trace link on the ID field — removes the custom data link that was causing a popup instead of direct navigation. - Added comprehensive X-Ray instrumentation guides for all 13 monitored services, including ready-to-paste Kiro prompts for each team to generate their own instrumentation specs.	1.5	
		Composite Alarm Docs, Noise Reduction Specs & Team Requests	- Documented all 6 active composite alarms with naming conventions, suppression behavior, and configuration examples. - Added 4 alarm noise reduction specs: notification grouping (platform digest emails), progressive backoff, quiet hours for maintenance windows, and alarm acknowledgement with DynamoDB TTL. - Renamed xray-tracing-team-requests.md to team-requests.md as a general team requests document. - Added log group standardization section with real naming inconsistency examples from AWS. - Combined services monitored table and first-action guidance into a single unified table in the alarms Confluence doc. - Expanded structured logging rationale with comparison table, structlog/Powertools code examples, and required SRE observability fields.	1.5	
		Log Group Observability Spec & Saved Queries Script	- Added log group observability spec covering 5 phases: discoverer, API endpoint, alarm email log links, Grafana CloudWatch Logs panel, and CloudWatch saved queries per platform. - Built saved queries script creating 30 CloudWatch Logs Insights saved queries per platform (errors, Lambda cold starts, timeouts, OOM, ECS task failures, OpenSearch slow queries). - Added log group URL utility for CloudWatch Console deep links, service type classification, and alarm-to-log-group matching.	1.5	
		Log Group Discovery Spec	- Added log group discovery spec to inventory repo covering the new LogGroupDiscoverer and /platforms/{platform}/log-groups API endpoint.	1.5	
		OpenAPI Title Fix	- Updated OpenAPI title assertion test to match renamed infrastructure-inventory service.	2.25	

		Meetings & Standups	Meetings this week (9 meetings, 2.5h): - 1:1 with Manager (Deotale) (30 min) - SRE Extended Review (60 min) - SRE Extended Review (60 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min) - SRE Weekly Call (30 min) - Daily Standup - Innovation IT (30 min) - Innovation IT - AWS Centralized Dashboard Planning (30 min) - Daily Standup - Innovation IT (30 min)	2.5	
		RPA Production Support Shift	- Reviewed APAC handover log and opened the SD Portal queue for the start of Week 20 rotation. - FX Closeout SD# 15285857 and SD# 15285410: Handled FX Closeout processing requests. Checked UiPath PortfolioDataops/DATAOPS-FX-CLOSEOUT folder for each client. Verified Daily565 folder — found old rep - FX Closeout PICTET_RPA failure (Processing_20260726_FAILURE, PICTET - SKIP LAMP): Checked UiPath folder for the PICTET_RPA client. Deleted last failed run. Checked Daily565 for old reports — zipped an - SD# 15109846: Status — awaiting end-user response since 2026-07-16. Closed per the IA Production Support guide. - Checked VPL queue at 5:00 PM EST — all items processed. - Updated Confluence shift handover log with all ticket statuses and open items. - Reviewed APAC handover log and SD Portal queue. - SD# 15285347 — AUTO REC DOWNLOAD Process, ASPECT client column update: Checked the \\uipath-share.int.coreprod01.citcosvc.com\UiPathShare\prod\CITCORECS_AUTO_REC_DOWNLOAD\SCRIPT\Rec_Download_Auto loca - SD# 15294806 — MESO cash completeness delay (Johansen): Checked MESO DB for pending items as requested. Checked CloudWatch Logs for errors. Informed Johansen on the status — cash-completeness was taki - Monitored VPL queue at 5:00 PM EST — queue clear. - Updated Confluence shift handover log. - Reviewed APAC handover log and SD Portal queue; followed up on SD# 15285347 (ASPECT client) — awaiting confirmation from Harsha on column updates.		

Week 20	Jul 27 – Jul 31, 2026	Development Work During Support Week	- Created new <code>ia_meso_process</code> repository for distributed tracing work — added Kiro steering files for project conventions (IISS-837). - Added Lambda Powertools X-Ray tracing spec (IISS-838) — defines approach for Lambda function distributed tracing using <code>aws-lambda-powertools[tracer]</code>. - Added EKS ADOT DaemonSet tracing spec (IISS-840) — covers OpenTelemetry-based tracing for EKS workloads. - Added API Gateway X-Ray tracing spec (IISS-841) — defines trace propagation at the API layer. - Added MSK/Kafka trace propagation spec (IISS-843) — covers distributed tracing across message broker boundaries. - Added EC2 standalone instance tracing spec (IISS-844) — defines tracing for non-containerised EC2 workloads. - Added <code>temp/</code> to <code>.gitignore</code> to prevent local working files from being accidentally committed. - Merged <code>feature/IISS-487-log-group-discovery</code> — includes composite alarms, saved queries script, and log group URL utilities built on July 24. - Tagged and merged SRE Observability Platform v2.19.9. - Added steering rule requiring descriptive messages for all git tags.	40	40
		<code>ia_meso_process</code> X-Ray Tracing	- Created new <code>ia_meso_process</code> repository for distributed tracing work — added Kiro steering files for project conventions (IISS-837). - Added Lambda Powertools X-Ray tracing spec (IISS-838) — defines approach for Lambda function distributed tracing using <code>aws-lambda-powertools[tracer]</code>. - Added EKS ADOT DaemonSet tracing spec (IISS-840) — covers OpenTelemetry-based tracing for EKS workloads. - Added API Gateway X-Ray tracing spec (IISS-841) — defines trace propagation at the API layer. - Added MSK/Kafka trace propagation spec (IISS-843) — covers distributed tracing across message broker boundaries. - Added EC2 standalone instance tracing spec (IISS-844) — defines tracing for non-containerised EC2 workloads.		
		Log Group Discovery Feature Merge	- Added <code>temp/</code> to <code>.gitignore</code> to prevent local working files from being accidentally committed. - Merged <code>feature/IISS-487-log-group-discovery</code> — includes composite alarms, saved queries script, and log group URL utilities built on July 24.		
		Development & Implementation	- Continued development work and implementation tasks.	8	
		Knowledge Base Organisation & Documentation	- Added comprehensive dev JIRA summary, timeline, and incident report to the knowledge base. - Added <code>.kiro/settings</code> to <code>gitignore</code> for clean repository hygiene. - Organised knowledge base structure with onboarding, infra, dashboard, and RPA support sections.	8	
		LLM Agent Critical Bug Fixes, Memory Increase & SSL/Timeout Reliability	- Added FastAPI improvements documentation and modern Python project structure migration plan. - Rewrote README to reflect current FastAPI application state. - Resolved 11 critical bugs across 6 Lambda functions — merged as <code>fix/critical-bug-fixes</code> into <code>release/v1.1.0</code>. - Increased <code>event_service</code> Lambda memory from default to 256MB to resolve OOM errors. - Added SSL verification helper and request timeouts to <code>event_service</code> Lambda. - Added timeout for Databricks API calls in <code>event_store</code> module. - Scoped SSL warning suppression to specific requests only in <code>ircoekb</code> module. - Added <code>event-service-reliability-fixes</code> spec and Python <code>.gitignore</code>.	2.75	

Week 21	Aug 03 – Aug 07, 2026	S3 Rename, Service Name Standardisation & Spec Cleanup	- Renamed S3 bucket from resources-monitoring to infrastructure-monitoring; updated repo name to ia-it-sre-infrastructure-monitoring-amg. - Cleaned up stale meridian- stack names; deleted old row_10_ecs_apis S3 objects. - Audited and tracked .kiro/specs in git.	2.75	41.5
		S3 Rename & Initial Spec Audit	- Renamed S3 bucket; moved Flask to dev dependencies; audited spec task completion.	2.75	
		Mapper Stack Cleanup, Service Rename & .env Cleanup	- Deleted resource mapper stack and all related stale scripts/docs after migrating to inventory API. - Marked superseded specs; renamed dashboard title and service/app name to ia-it-sre-infrastructure-monitoring-amg. - Cleaned up .env: removed redundant variables, switched Grafana state reads from .env to AWS (Secrets Manager/SSM), removed APP_ENV. - Updated AWS_CA_BUNDLE guidance to use Windows user environment variable.	1.25	
		Spec Closures, CW Metrics Live, IAM Fix & Retire Mappers	- Marked specs complete: ECS target group discovery Phase 5, CloudFront S3 proxy phases 1-3, lambda-api-migration tasks N/A. - Added retire-mappers spec; Phase 4 complete — mapper stack deleted. - Enabled PUBLISH_CW_METRICS=true — CloudWatch metrics publishing verified live in DEV. - Fixed elasticloadbalancing IAM permissions; renamed stack to infrastructure-inventory.	1.25	
		CloudWatch Log Group Discoverer (row_16)	- Added CloudWatch Log Group discoverer (row_16) to the infrastructure inventory API. - Merged inventory-tasks feature branch covering cache guard improvements, log-group discovery, and spec audits.	1.25	
		API Stack Rename Spec & Implementation	- Added api-stack-rename spec with detailed task list. - Renamed inventory API CloudFormation stack to infrastructure-inventory naming convention. - Added force-update mechanism for Lambda deployments.	1.25	
		LLM Agent Major Refactor	- Added pyproject.toml, requirements/ folder, .env.example; archived dead code; externalised WIKI_LIBRARIES to S3 JSON config. - Added Kiro hooks and quality gates to buildspec; added idempotent KB sync with SharePoint Modified timestamp detection (Phase 1). - Migrated all Lambda modules to structlog; added Pydantic CfnEvent and AgentEvent models. - Added type hints and Google docstrings across all Lambda modules. - Refactored 1140-line event_service/function.py into 8 focused modules; removed sys.path.insert hacks. - Added 146 unit tests; iteratively improved coverage 60% → 88% → 96% → 99% (194 tests) — v1.1.24 through v1.1.32. - Renamed secrets.py → event_service_secrets.py; fixed 96 ruff lint issues automatically.	1.25	
		ECS Task Allocation Prod Alignment	- Fixed ECS task-level CPU and memory allocation to match container limits (512 CPU / 1024MB) and align exactly with production. Bumped to v1.1.55.	1.25	

		handler_utils Decorators, SRP Refactor, CFN Nested Stack & 233 Tests	- Reached 199 tests with 100% coverage; built shared handler_utils with log_handler, scheduled_handler, cfn_handler decorators — applied to all Lambda handlers (v1.1.35, 217 tests). - Grew test suite: 199 → 217 → 225 → 233 tests through SRP refactors, type hint completion, CFN response coverage, and ircoekb SRP refactor (v1.1.35–v1.1.39). - Fixed coverage config; caught UnauthorizedSSOTokenError early; archived 5 dead scripts; moved data files to data/. - Synced all docs to v1.1.40; added BACKLOG.md (8 deferred items), decisions/ folder, Lambda function name reference table. - Applied CFN tier-1 and tier-2 improvements; built parallel structured nested CFN stack (v1.1.46). - Fixed Bandit security issues; resolved all pre-commit hook failures (v1.1.47). - Added env prefix to Lambda names then reverted to avoid breaking production (v1.1.48–v1.1.49). - Standardised .yaml → .yml; removed incorrect S3 object expiry rules to retain KB content indefinitely (v1.1.52–v1.1.53).	4	
		CW Metrics Spec Complete & Cache Guard	- Updated all four inventory task specs with accurate current state. - Marked cloudwatch-metrics-publishing ALL PHASES COMPLETE. - Added API-refresh guard to prevent 0-resource cache overwrites.	4.25	
		Meetings & Standups	Meetings this week (5 meetings, 1.5h): - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min)	1.5	
		KB Content Upload Script, Bedrock Agent Instruction Externalisation &	- Added kb-content/ folder structure and upload_kb_content.py script for manually uploading NAV Checklist PDFs to the Bedrock Knowledge Base S3 bucket (v1.1.56). - Updated kb-content folder to mirror Confluence hierarchy with faq/ subfolder; upload script now recurses subfolders. - Added --env flag (dev/uat/prod) with production confirmation prompt to prevent accidental uploads to production. - Added stdlib logging and 36 unit tests to the upload script; configured scripts/ in the pytest pythonpath. - Fixed KB sync Lambda fallback name — actual DEV function has no env prefix. - Added ServerSideEncryption=aws:kms to all S3 put_object calls to comply with CITCO-S3ForceEncryptNoPublic policy. - Refactored upload_kb_content.py into _kb_utils/ package with separate auth, s3, sync, and bucket modules — added 48 per-module unit tests. - Fixed KB sync to always use us-east-1 for sync Lambda regardless of S3 upload region.	4	
		API Stack Rename & Force-Update Lambda	- Renamed inventory API CloudFormation stack to infrastructure-inventory naming convention. - Added force-update mechanism for Lambda deployments to ensure code changes are always picked up on deploy.	4	

Week 22	Aug 10 – Aug 14, 2026	Buildspec YAML Fixes, KB Upload CLI Tests & Critical Bug Fixes	- Added 30 unit tests for upload_kb_content.py CLI covering _get_kb_ids, cmd_status, cmd_upload, and main functions. - Fixed multiple YAML_FILE_ERROR issues in buildspec.yml — replaced unicode box-drawing characters with ASCII, quoted echo commands containing colons, and replaced \${COMMIT_HASH:=latest} with if/else block. - Added quality gates to buildspec and fixed critical broken paths. - Resolved 11 critical bugs across 6 Lambda functions. - Cherry-picked KB upload improvements: Confluence hierarchy mirroring, --env flag with prod confirmation, stdlib logging + 36 tests, KB sync fallback name, aws:kms SSE, _kb_utils/ package refactor (48 tests), us-east-1 sy - Reverted buildspec.yml to PR 3015 state after multiple failed fix iterations, then applied clean hotfix for update_json.py path (functions/ → scripts/).	8	41.5
		Buildspec cp Hotfix, Safe Dict Access & Bedrock Guardrail Streaming Fi	- Fixed buildspec to add cp commands packaging cfn_response.py into Lambda deployment zips — upload, Athena, and OpenSearch functions were failing with ModuleNotFoundError without it. - Fixed unsafe dictionary key accesses in event_service — replaced direct key lookups with .get() and try/except guards. - Created shared dict_utils.safe_get utility and replaced all unsafe dict accesses across function.py. - Replaced %s logger format strings with f-strings throughout function.py and dict_utils.py. - Expanded Bedrock agent SCOPE instruction to correctly handle follow-up messages and conversational references. - Fixed streaming path to detect Bedrock guardrail INTERVENED trace events and replace the blocked chunk with a clean out-of-scope message instead of showing raw BlockedOutputsMessaging mid-response. - Improved streaming guardrail handling to buffer all chunks before sending — if a guardrail fires at any point, the entire partial response is discarded and a clean message is sent instead.	8	
		ECR Image MVC Refactor, Local Dev Auth & Guardrail Fixes	- Added LOCAL_DEV_USER environment variable to bypass AD authentication for local testing. - Added python-dotenv to auto-load .env files — developers no longer need to manually source environment variables. - Fixed guardrail INTERVENED handling in non-streaming path — was returning empty content, now returns a clean out-of-scope message. - Fixed non-streaming path chunk key from event['bytes'] to event['chunk']['bytes']. - Completed full MVC refactor through three iterations — src/app/ package → root app/ package → full MVC structure with structlog, pydantic-settings, and proper package layout.	4	
		ircoekb SharePoint Timestamp Sync & BuildSpec CodeBuild Fixes	- Added hash-based idempotent sync for ircoekb KB pages; upgraded to SharePoint Modified timestamp detection (Phase 1). - Added ircoekb sync justification with 93-day evidence and retry pattern analysis; moved ircoekb_config.json from scripts/ to data/. - Fixed buildspec to upload ircoekb_config.json to S3 on every build. - Resolved series of CodeBuild pipeline issues: AWS_ACCOUNT_ID not a built-in variable (fixed via STS), then via DATA_BUCKET_NAME env var (CodeBuild runs in CICD account), then via CODEBUILD_BUILD_ARN, then fixed YAML pars - Final clean solution: read app account ID from parameters/{env}.json CODLAccountld. - Removed cp commands for non-existent shared modules. - Reverted a direct push to main that bypassed the PR process.	4	

		BuildSpec Pipeline Fixes, ircoekb Local Config & CFN Template Fixes	- Fixed series of buildspec pipeline issues: account ID extraction from ModelName ARN, S3 upload permission gaps (CodeBuild role lacks permissions), and progressively commented out cfn-lint, ruff, bandit, and unit test qua - Fixed duplicate BUCKET_NAME environment variable in event_service Lambda CloudFormation template. - Merged buildspec quality gates hotfix into main. - Implemented idempotent sync with local config file for ircoekb knowledge base integration. - Fixed YAML template issues and resolved ECSService integer-as-string CloudFormation validation errors.	4	
		ECR Image MVC Refactor, Docker Hardening, LDAP/CORS Security & Test Su	- Brought in full MVC refactor: app/ package structure with structlog and pydantic-settings. - Added improvement plan for v2.0 release. - Fixed CFN template validation errors in cfn.lambda.yml. - Hardened Docker container image for the LLM ECR service. - Applied LDAP TLS and CORS lockdown security fixes. - Applied code quality improvements. - Added full test suite with uv project setup. - Added .python-version tracking to git following the uv convention.	4	
		Meetings & Standups	Meetings this week (5 meetings, 1.5h): - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min)	1.5	
		MVC Refactor Complete, 280 Tests 100% Coverage & ircoekb Fixes	- Bumped ircoekb Lambda timeout from 300s to 900s to handle longer processing runs. - Updated ircoekb improvements documentation with current status and S3 HEAD optimisation. - Completed full repo refactor with unit tests from feature branch; added quality gates. - Applied SSL hotfix to skip certificate verification for internal Citco SDM endpoints; added nossec B501 annotations. - Archived upload_ircoekb_config.py and consolidated all archived content into _archive/. - Fixed 3 buildspec issues found by validate_buildspec.py; added 14 unit tests for it. - Added scripts/ to coverage reporting in buildspec. - Progressively improved coverage 60% → 99% → 100% with 234 → 247 → 280 tests.	2	
		ECS ADOT Sidecar X-Ray Tracing	- Added spec for ECS ADOT sidecar X-Ray tracing (IISS-839). - Added ECS ADOT sidecar CloudFormation snippet and X-Ray config. - Wired X-Ray config into config_manager and run.py; merged into release.	2	
		SQS Trace Header Propagation	- Added spec for SQS trace propagation (IISS-842). - Implemented X-Ray trace header propagation to SQS producers. - Added improvements backlog (6 items) and documented structlog migration strategy for log call standardisation.	2	
		ia_infrastructure_dashboard Initial Commit	- Created initial commit for the new ia_infrastructure_dashboard BitBucket repository — the Enterprise Custom UI / Infrastructure Monitoring Dashboard (IISS-919).	2	

Week 23	Aug 17 – Aug 21, 2026	Agent Instruction Externalisation, Usage Report Redesign, CFN Nested S	- Externalised Bedrock Agent instruction to agent_instruction.txt in agent/ folder; fixed builds spec to upload via UploadFileFunction (no cross-account S3). - Replaced .format() calls with f-strings; migrated upload and opensearch Lambda packages to structlog. - Added README.md to all Lambda packages and shared/; fixed security issue replacing eval() with json.loads(). - Completely redesigned usage email report as a pivot table with summary cards, Citco brand colours (#002D72 navy), Aptos styling, and HTML templates extracted to templates/. - Fixed 32 cfn-lint warnings and applied security hardening across CloudFormation templates. - Built comprehensive structured nested CFN stack: cfn.storage, cfn.iam, cfn.lambda, cfn.bedrock, cfn.ecs, cfn.monitoring orchestrated by cfn.main. - Added SourceAccount to KBIgestionSchedulePermission for LAMBDA_FUNCTION_PUBLIC_ACCESS_PROHIBITED compliance. - Cleaned up repository: removed stale dependencies/, redundant confest.py, generated HTML reports, and leftover PR docs.	8	41.5
		Lambda Powertools X-Ray Tracer Instrumentation	- Created spec for Lambda Powertools X-Ray Tracer instrumentation. - Added @tracer.capture_lambda_handler and @tracer.capture_method decorators to all Lambda functions. - Fixed @tracer.capture_method on wait_for_query and fixed cold-start S3 data loading to work with the capture decorator. - Added tracing.md documenting all decorated and undecorated functions with rationale.	2	
		Docs Diagrams, ircoekb Hybrid Sync, Pydantic Models, Kill Switch, Retr	- Upgraded all plain-text docs to box-drawing ASCII art with consistent 2-space padding; added Mermaid diagrams (neo dark theme) to architecture, chunking, and vector-search docs. - Added script to fetch IRCOEKB production logs; collected PROD summary data for Aug 15-19. - Added detailed specs for IRCOEKB timeout and 500 error fixes; added probe script to test SharePoint ALB filter support. - Implemented hybrid two-phase IRCOEKB sync fix — timeout 300s → 600s with parallel processing. - Added Pydantic validation models for BPM API, IRCOEKB config, Databricks, and wiki sync. - Added 5 new IA Best Practice improvement specs; marked completed tasks across 13 spec files. - Implemented SSM-based process kill switch for IRCOEKB, mapping, and event_store with operations guide. - Added tenacity exponential backoff retry logic to all external API calls.	2	
		Dedicated Log Groups API Endpoint	- Added dedicated GET /api/log-groups endpoint to the inventory API — serves log group data for the alarm email log links and Grafana CloudWatch Logs panel (IISS-845 Phase 2).	2	
		ECS ADOT Sidecar Instrumentation Spec	- Added ECS ADOT sidecar instrumentation spec and task list to the ECR image repo.	2	
		Docs Overhaul, Composite Re-Notification & Citco SES HTML Emails	- Completed comprehensive documentation overhaul — updated all docs to reflect current repo state, fixed inventory row numbers, added ECR/EKS/ECS Target Groups inventory docs, removed duplicates, renamed folders. - Decommissioned mapper Lambda references from docs; added Mermaid architecture diagrams with consistent dark theme. - Corrected alarm counts throughout documentation. - Built composite alarm re-notification with combined email at 5-minute cadence for repeated alerting. - Built Citco-branded HTML alarm emails via SES with per-platform digest; verified bbhagat@citco.com as DEV SES sender.	2	

		Docs Mermaid Theme & Spec Closures	- Marked completed specs: cfn-restructure, cli-class-refactor, flask-api-cleanup. - Removed stale env example files; updated .env.dev.example. - Applied consistent neo dark theme to all Mermaid diagrams; fixed subgraph background colour (pink → dark navy).	2	
		@citco/core + @citco/theme Plan	- Added IISS-919 plan to replace shadcn/ui + custom citco-ui with @citco/core + @citco/theme for the Enterprise Infrastructure Monitoring Dashboard.	2	
		SES Failure Notifications, 99% Coverage, Pinned Deps & IA Best Practic	- Added SES failure/status notifications for IRCOEKB and mapping Lambdas with Citco-styled HTML emails and clickable CloudWatch Logs deep-links. - Achieved 99% test coverage across all meaningful files including scripts/. - Migrated all requirements/.txt into pyproject.toml and pinned all dependency versions; also pinned Lambda requirements.txt to exact versions. - Implemented IA Best Practice improvements: PII masking, health checks, rerun validation, exception classification, S3 encryption, and actionable failure alerts. - Added manual fallback guide and populated all ia-project-manifest.json links. - Resolved all deptry dependency issues; added deptry to pre-commit hooks. - Added IA Best Practice verification reports for Aug 5, Aug 19, and Aug 20.	2	
		SMTP Email Delivery, Notifications CLI, Project Rename & AWS Icons	- Merged three feature branches into release and then into main — mapper decommission, composite alarm re-notification, Citco HTML emails. - Fixed architecture Mermaid diagram Services subgraph layout to a cleaner 4x2 grid. - Switched alarm email delivery from SES to file-based HTML templates with SMTP via IA shared credentials (IASecrets/TOIA-SMTP-Credentials). - Fixed deploy script to include email_templates/.html in alarm Lambda zip packages. - Fixed IAM: added missing Resource on cloudwatch:DescribeAlarms, scoped Secrets Manager ARN, removed stale SES permissions, added kms:Decrypt for SMTP credentials KMS key to both alarm Lambda roles. - Added notifications CLI subcommand to start/stop/check status of alarm Lambda schedules; refactored alarm_notifier stop/start to use SNS unsubscribe/subscribe instead of Lambda concurrency. - Documented notification controls CLI with --stop behaviour, health metrics impact, and UAT/PROD environment examples. - Renamed project from IA IT SRE Platform Monitoring to IA IT SRE Infrastructure Monitoring across all HTML templates and specs.	4	
		ia-bpo Platform Grouping & CI/CD Pipeline Spec	- Added ia-it-sre resources under the ia-bpo platform in the inventory API; updated platform config documentation. - Removed stale force-update workaround note. - Added CI/CD pipeline deployment spec for the DevOps team; promoted LambdaCodeVersion from optional to required in the spec.	4	
		Meetings & Standups	Meetings this week (5 meetings, 1.5h): - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min) - Daily Standup - Innovation IT (30 min)	1.5	

Week 24	Aug 24 – Aug 28, 2026	RPA Production Support Shift	- Reviewed APAC handover log and opened SD Portal queue for Week 24 rotation start. - Monitored UiPath Orchestrator for overnight queue items across all active processes. - Triaged incoming tickets; confirmed successful overnight runs for MESO and FX Closeout. - Checked VPL queue at 5:00 PM EST — all items processed. - Updated Confluence shift handover log with end-of-day summary. - Monitored SD Portal queue and Teams channels throughout shift. - Handled MESO process status inquiries via CloudWatch Logs run ID trace. - Investigated FX Closeout retrigger — verified Daily565 folder, triggered in UiPath. - Reviewed APAC handover and coordinated on pending escalations. - Updated Confluence shift handover log. - Monitored SD Portal queue and reviewed APAC handover log. - Handled Auto Rec Download config request — backed up and updated gridview.py.	40	40
Week 25	Aug 31 – Sep 04, 2026	Final KT Handover & Infrastructure Monitoring Wrap-up	- Conducted final knowledge transfer session with Horace and Kri on dashboard operating model and alarm runbooks. - Documented incident escalation procedures and handover notes for team continuity. - Confirmed Mitacs BSI renewal scope for Sep–Dec 2026: X-Ray tracing, log-group discovery, and IISS-919 frontend. - Reviewed IISS-919 infrastructure monitoring dashboard final state with Horace. - Participated in final Innovation IT standup and team handover session.	8	8
TOTAL INTERNSHIP HOURS				900	900